9 Obturation of the Cleaned and Shaped Root Canal System

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Importance of Effectively Sealing the Root Canal System

The authors of a meta-analysis of factors influencing the efficacy of nonsurgical root canal treatment (NS RCT) reported that the following four factors influenced success: the absence of a periapical lesion (PL), root canal obturations containing no radiographic voids, obturation to within 2.0 mm of the radiographic apex, and an adequate coronal restoration.[196] On the basis of the current best available evidence, both the quality of the endodontic treatment and the coronal restoration are significant in any healing outcome.[104]

In a recent controlled animal study, evidence suggests that cleaning and shaping (C&S) of the radicular space is more important than the method of obturation.[240] PLs were created by removing the pulp and leaving the teeth open to the oral cavity. In the control group, the canals were cleaned and shaped before obturation with gutta-percha (GP) and a resin-based sealer. The teeth of the experimental group were cleaned and shaped as in the control group but left unobturated. At 190 days, the animals were killed and histological evaluations were performed. There was no difference in the healing between the instrumented and obturated teeth and the instrumented and unobturated teeth.

To date, there is no effective method for determining whether C&S procedures have been effective.[10,11] There is some evidence that the criteria of clean dentinal filings and/or enlargement beyond the first file to bind at the working length have proven to result in a healing outcome.[244] Although the length of preparation has been emphasized, the irregular canal diameter may be a more significant factor in healing or nonhealing outcomes.[133] Evidence indicates root canal systems (RCS) are often underprepared in the apical one third of the RCS.[56] Historically, culturing has been employed and obturation delayed until a "negative"

culture was obtained. In contemporary endodontic treatment, culturing has been abandoned during routine care.[250] When dealing with vital pulp tissue, preventing bacterial contamination is a primary concern. In RCS with a necrotic pulp, the organisms involved in the disease process are primarily facultative or obligate anaerobes that may not grow in culture. Molecular microbiologic techniques (polymerase chain reaction) have demonstrated that a variety of organisms are present that do not grow in culture.[13,237,273] The role these organisms play in the disease process is not well understood.[189] The reader is referred to Chapters 15 and 16 for a fuller discussion.

The process of C&S determines both the degree of disinfection and the ability to obturate the radicular space. Obturation is therefore a reflection of the C&S and is evaluated on the basis of length, taper, density, level of GP obturation, and the coronal seal, that is, an adequate provisional or definitive restoration (Fig. 9.1). It is not possible to assess the quality of the seal established during obturation with a radiograph, and it is important to recognize that no material or technique prevents leakage.[4,58,81,139,182,209,210] Indeed, obtaining an impervious seal may not be feasible because of the porous tubular structure of dentin[4] and canal irregularities.

The primary etiology of pulpal and periradicular pathosis is, as discussed in Chapter 16, microbiological.[141,181,186,239] Pulpal remnants, necrotic tissues, bacteria, and bacterial by-products remaining in the inaccessible areas of a C&S RCS can initiate and/or perpetuate a PL because the host defense mechanisms are unable to remove them. Studies indicate that an RCS cannot be completely cleaned and disinfected.[10,11,122] Obturation of the radicular space is necessary to minimize coronal leakage from the oral environment. Obturation reduces coronal leakage and bacterial contamination, seals the apex from the periapical tissue fluids, and entombs the remaining irritants in the canal.

Coronal leakage has been proposed to contribute to treatment failure based on in vitro leakage studies.[293,294] The clinical implication is that retreatment has to be performed in those teeth that are not restored definitively 3 months after NS RCT. The clinical significance of this issue has been challenged.[232,233] A recent systematic review and meta-analysis of the results derived from nine similar studies indicate that poor-quality NS RCT and poor-quality coronal restorations have similar odds in adversely affecting the healing of apical periodontitis.[104] Based on the best available current evidence, the odds for healing apical periodontitis increase with both adequate NS RCT and an adequate restorative treatment.[104] It must be emphasized that these two factors were examined in isolation in those studies. These postoperative prognostic factors, along with other preoperative and intraoperative prognostic factors, have been reported to contribute significantly to periapical healing in primary and secondary NS RCT.[194] Nevertheless, maintaining an effective coronal seal through optimal obturation of the instrumented RCS and placing an appropriate restoration, as discussed in Chapter 23, should be considered essential components of a positive healing outcome for NS RCT.[124,132]

Clinicians are concerned with whether it is more appropriate to place a definitive restorative material instead of a

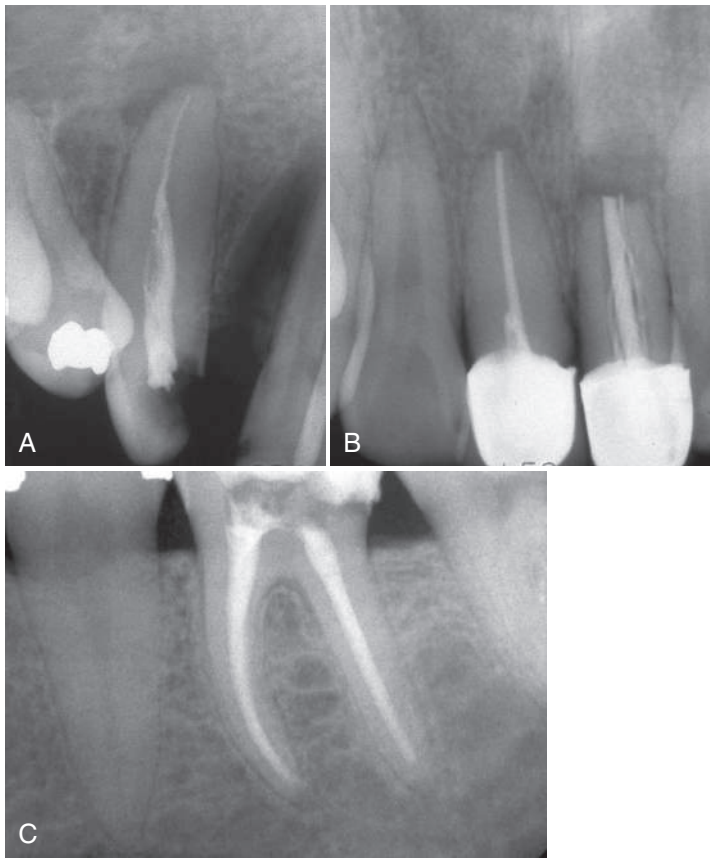


Fig. 9.1 Examples of inadequate obturation. **A,** Maxillary right canine with adequate length but lacking density and no coronal seal. Central incisor is filled to adequate length, but obturation exhibits voids. **B,** Maxillary central incisors. Maxillary right central incisor exhibits a lack of density and taper. Maxillary left central incisor has voids and unfilled canal space. **C,** Mandibular left first molar with adequate obturation; provisional restoration shows poor adaptation on the distal because of the failure to remove caries.

temporary material to prevent leakage.[169,295] This is a controversial issue, as previous studies failed to provide definitive evidence to demonstrate that the presence of a definitive restoration will contribute to a long-term healing outcome for NS RCT when those RCS are optimally obturated.[62,243] These results have been supported by a more recent prospective clinical study that examined the factors affecting the outcomes of NS RCT.[194] In this study, the authors reported that the type of coronal restoration had no significant influence on periapical healing as long as those restorations were of good quality. Paradoxically, the same group of authors reported in a separate study that cast restorations, when compared with temporary restorations, significantly improved the survival of teeth that had undergone primary or secondary NS RCT.[194] This may be due to protection of remaining, weakened tooth structure by full-coverage cast restorations that enabled those teeth to function longer. Regarding the necessity for retreatment of root-filled teeth without definitive restorations, there is no clear evidence to justify that retreatment has to be performed in teeth in which a temporary restoration had been inserted for more than 3 months solely because of the suspicion of microleakage.[5,144] One may consider replacing the temporary restoration with a definitive coronal restoration immediately and to observe the tooth for at least 3 months before placing a full-coverage restoration.[144]

Three-dimensional obturation of the RCS is essential to a long-term healing outcome. The RCS should be sealed apically, coronally, and laterally. Various methods have been advocated for obturation. Unfortunately, all materials and techniques result in some degree of leakage.[314] Although a poorly obturated RCS and leakage are correlated, there are no standardized techniques to evaluate and measure microleakage.[134] Also, radiographic evaluation of obturation does not correlate well with leakage.[119] An adequate two-dimensional radiographic appearance of the obturation may not correlate with an adequate seal (Fig. 9.2).[119] Variation in radiographic interpretation by the clinician, the overlying osseous structures, and the lack of uniformity in the obturation materials are significant variables.[83,84,146,290]

The diagnostic outcome of NS RCT is based on clinical and radiographic findings. In a series of prospective studies, the Toronto group evaluated the success and failure of NS RCT at 4 to 6 years after completion of treatment. For primary NS RCT,[72,93,97,177] teeth with preexisting apical periodontitis were found to have a lower healing rate (82%) compared with teeth without a periapical radiolucency (93%). Better outcomes were associated with teeth without a periapical radiolucency, with single roots, and without midtreatment complications, such as a root perforation. Teeth were treated by using a flared preparation and vertical compaction of warm GP or step-back preparation and lateral compaction of GP. Differences were noted with the adequacy of the obturation and the treatment technique. Adequate length had a higher healing rate (87%) when compared with inadequate length (77%). The flared preparation and vertical compaction had a higher success rate (90%) when compared with a step-back preparation and lateral compaction (80%). For retreatment NS RCT,[73,92] teeth with preexisting apical periodontitis were also found to have a lower healing rate (80%) compared with teeth without a periapical radiolucency (93%). Better outcomes were achieved in teeth with an inadequate previous obturation, without perforation and a radiolucency. Similar results were reported in a more recent prospective clinical study that examined the effects of primary or secondary NS RCT on periapical healing.[194] The percentage of roots with complete periapical healing after primary (83%) or retreatment (80%) NS RCT was similar. Absence of preoperative apical periodontitis was identified as one of the prognostic factors affecting primary or secondary NS RCT. In the presence of apical periodontitis, treatment prognosis was significantly improved by the presence of a smaller PL. These studies agree with earlier work[275,282] indicating preexisting apical pathosis as a major factor reducing a favorable prognosis and highlighted obturation technique as a factor influencing healing or nonhealing outcomes.

Although periapical radiographs have been used to examine the healing of posttreatment apical periodontitis since 1922,[195] their suitability for evaluating the outcomes

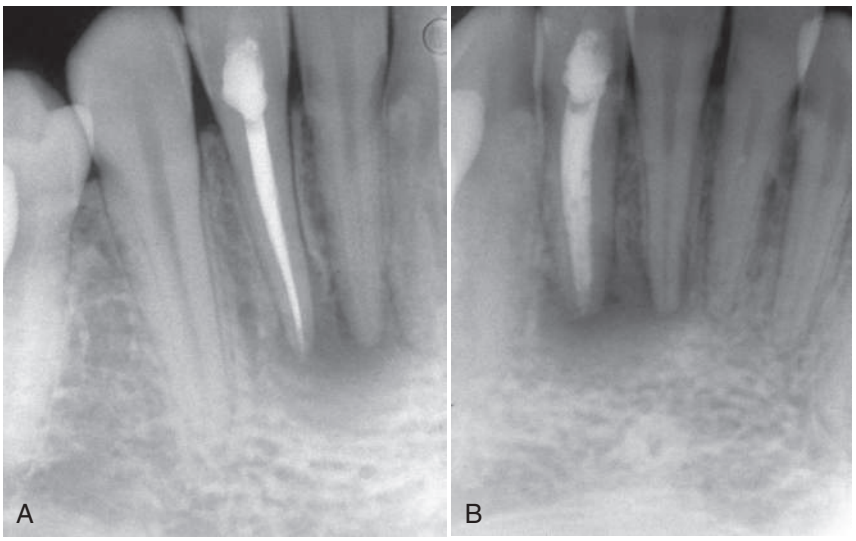


Fig. 9.2 A, Posttreatment radiograph of a mandibular left lateral incisor with ostensibly adequate obturation. **B,** Angled view reveals voids.

of NS RCT has recently been challenged, in that previously reported outcomes based on this assessment modality may have overestimated the healing rates of apical periodontitis.[12,313] With the advent of cone-beam computed tomography (CBCT), a better diagnosis of PL within the cancellous bone can be made. This is because CBCT creates reconstructed images from slices of data in any plane and location of the volume of interest, thus eliminating the lack of three-dimensional assessment and anatomical noise, which hampers the accuracy of periapical radiography. This results in a higher signal-to-noise ratio and image contrast, and improves the detection of periapical radiolucencies.[217,253] Although intraoral radiographs are the imaging modality of choice, when two-dimensional intraoral radiography is inconclusive, the use of CBCT imaging was reported to have twice the odds of detecting periapical pathosis than traditional periapical radiography.[12] Though different expectations in endodontic outcomes provide insight into the definition of the success of these procedures, it remains questionable whether this gain of additional information with CBCT in endodontic outcomes would lead to a better healing outcome in endodontic treatment. The results of these studies emphasized that developing good technical skills for obturating an RCS should not be underscored and that suitable canal obturation methods should be developed to achieve more compacted, void-free filling materials and at the correct length.

Historical Perspectives

Over the past 70 to 80 years, the dental community has seen attempts to improve the nature of root canal obturation with cements/sealers and with variations in the delivery of GP to the prepared RCS. During this era, the impetus for these developments was based heavily on the continued belief in the concept of focal infection, elective localization, the hollow-tube theory, and the concept that the primary cause for failure of NS RCT was the apical percolation of fluids, and microorganisms into a poorly obturated RCS.[81] Despite endeavors reporting an adequate RCS obturation radiographically, a "perfect" seal is not achievable, as most current materials have no antimicrobial activity and do not seal the RCS.[47] From this chronological perspective of technical and scientific thought, this chapter clarifies and codifies contemporary concepts in the obturation of the C&S RCS.

Timing of Obturation

Factors influencing the appropriate time to obturate an RCS include the patient's signs and symptoms, status of the pulp and periradicular tissues, the degree of difficulty, and patient management.

VITAL PULP TISSUE

At present, the consensus is that one-step treatment procedures are acceptable when the patient exhibits a completely or partially vital pulp.[175] Removal of the normal or inflamed pulp tissue and performance of the procedure under aseptic conditions should result in a healing outcome because of the relative absence of bacterial contamination. Obturation at the initial visit also precludes contamination as a result of coronal leakage during the period between patient visits.

Elective NS RCT for restorative reasons can be completed in one visit provided the pulp is vital, to some degree, and time permits. Obturation of RCS in patients whose condition is urgent depends on the pretreatment diagnosis. When pain occurs as the result of irreversible pulpitis, obturation can occur at the initial visit because removal of the vital pulp tissue will generally resolve the patient's pain.

NECROTIC PULP TISSUE

Patients who present with pulp necrosis, with or without asymptomatic periradicular pathosis, for example, asymptomatic apical periodontitis, chronic apical abscess, or condensing osteitis, may be treated in one visit based on the best available information. When patients present with acute symptoms caused by pulp necrosis and acute periradicular abscess, obturation is generally delayed until the patient is asymptomatic. However, more than 30 years ago, investigators demonstrated that cases with soft-tissue swelling could be completed in one visit with appropriate endodontic treatment, incision for drainage, and a regimen of antibiotics.[277] Management of these patients, however, may be more difficult should problems persist or become worse after the completion of treatment.

During the 1970s, there was concern about the timing of obturation. Performing endodontic treatment in one visit was controversial. Conventional wisdom suggested that patients would have a higher incidence of posttreatment pain. However, clinical studies[185,213,298] and systematic reviews[175] indicated that there was no significant difference in the healing rates of apical periodontitis between single-visit and multiple-visit NS RCT. Patients experienced less frequency of short-term postobturation pain after single-visit than those receiving multiple-visit NS RCT.[281]

With the introduction of the concept of biofilms, other investigators examined the intracanal microbial status of 16 mesial roots of human mandibular first molars with primary apical periodontitis immediately after one-visit NS RCT.[190] In that study, the instrumented RCS were irrigated with 5.25% NaOCl and 17% ethylenediaminetetraacetic acid (EDTA), and obturated with GP and zinc oxide eugenol sealer. The apical portion of the root of each tooth was surgically removed and prepared for correlative light and transmission electron microscopy examination. Fourteen of the 16 root-treated teeth revealed residual intracanal infection after instrumentation, antimicrobial irrigation, and obturation. The microbes existed mostly as biofilms in inaccessible recesses of instrumented main canals, the intercanal isthmus, and accessory/lateral canals.

In a histobacteriological study, the in vivo microbiological status of the mesial roots of mandibular molars with primary apical periodontitis were examined after single-visit or two-visit NS RCT.[298] Those roots were instrumented and irrigated with 5% NaOCl, 17% EDTA, and 2% chlorhexidine. In the single-visit group, the RCS were immediately obturated, while in the two-visit group, calcium hydroxide ($Ca[OH]_2$) dressing was placed in the RCS for 1 week prior to obturation. In the single-visit group, no RCS, out of the six

RCS examined, was completely free of bacteria. Residual biofilms were identified in the main RCS, isthmus, apical ramifications, and dentinal tubules. In the two-visit group, two out of seven roots were rendered bacteria-free. Residual biofilms were found predominantly in the isthmus and ramifications, intermixed with necrotic tissues and debris. Taken together, these studies illustrate that microbes are extremely difficult to eliminate within the complex RCS and continued endeavors have been made to improve microbiological status. Current evidence points out that the use of 5% NaOCl and 17% EDTA is efficacious in reducing microbes.[194] When one appointment is not achievable, the use of $Ca(OH)_2$ has been advocated as it remains the best medicament available to reduce residual microbial flora.[157] Calcium hydroxide has been advocated as an antimicrobial and temporary dressing in necrotic cases that cannot be treated in one visit[274] because investigators noted that bacteria in instrumented, unfilled RCS can multiply and reach their pretreatment numbers in 2 to 4 days.[48] Nevertheless, the ability of $Ca(OH)_2$ to completely eradicate microbial species from the RCS has recently been questioned. In vitro studies demonstrated that the antibacterial activity of $Ca(OH)_2$ can be inactivated by dentin.[116] Other clinical studies reported that the number of bacterial positive RCS did not decrease after the use of $Ca(OH)_2$ as an interappointment dressing.[222] A systematic review and meta-analysis of eight clinical studies concluded that $Ca(OH)_2$ is useful but have limited effectiveness in completely eliminating bacteria from human RCS when evaluated by culture techniques.[249] Although $Ca(OH)_2$ has a wide range of antimicrobial activity against common endodontic pathogens and is an effective antiendotoxin agent, it is less effective against *Enterococcus faecalis* and *Candida albicans*.[91,184] Alternative interappointment dressings such as the use of chlorhexidine gel, $Ca(OH)_2$ in chlorhexidine gel, or triple antibiotic paste (metronidazole, minocycline, and ciprofloxacin) has been proposed, with dichotomous results.[15,75,203,272,302] Although there are clinical trials to support the use of $Ca(OH)_2$ or alternative interappointment dressings, they do not reduce microbes beyond levels already achieved by canal preparation with NaOCl.[172,176,271] It is fair to say that interappointment antimicrobial dressings are fairly well accepted by clinicians in cases that cannot be treated in a single visit.

In general, obturation can be performed after C&S procedures when the RCS can be dried and the patient is not experiencing significant pain and or swelling. An exception is the presence or persistence of exudation from the canal. Obturation of an RCS that cannot be dried is contraindicated. Procedural concerns also dictate the time of obturation. Difficult cases may require more time for preparation and can be managed more uneventfully in multiple appointments. Patients may require multiple short appointments because of medical conditions, their psychological state of mind, and/or fatigue.

Length of Obturation

One of the controversies in endodontics that remains unresolved is the apical limit of NS RCT and obturation.[194,231,255] Early studies identified the dentinocemental junction as the apical limit for obturation.[153] However, this histologic landmark cannot be determined clinically and it has been found to be irregular within the canal. For example, the dentinocemental junction may be several millimeters higher on the mesial canal wall when compared with the distal wall. Additionally, the dentinocemental junction does not coincide with the narrowest portion of the canal or apical constriction. The reader is referred to Chapter 7 for more information about this anatomy.

Traditionally, the apical point of termination has been approximately 1 mm from the radiographic apices as determined by radiographs. Kuttler[153] noted that the apical anatomy consists of the major diameter of the foramen and the minor diameter of the constriction (Fig. 9.3), with the apical constriction identified as the narrowest portion of the canal. The average distance from the foramen to the constriction was found to be 0.5 mm, with the foramen varying in distance from the apex up to 2.5 mm. Kuttler also noted that the foramen-to-constriction distance increases with age because of cementum deposition. Supporting this finding, other investigators reported that the location of the foramen was not at the apex. Deviations occurred in 92% of the roots and averaged 0.6 mm.[46] Another study noted the average apex-to-constriction distance was 0.9 mm and that 95% of the constrictions were between 0.5 and 1 mm in diameter.[85] This study also noted that the classic apical anatomy described by Kuttler was present in only 46% of teeth. Other variations identified were a tapering constriction, the multiconstriction, and the parallel constriction. Other investigators examined 230 roots of permanent teeth stereomicroscopically and with radiographs.[31] Results of this study indicated a deviation of the foramen from the apex in 76% of the roots with microscopy and 57% with radiography. The mean distance was 1 mm.

The authors of a scanning electron microscopic study reported that no foramina coincided with the longitudinal axis of the root with the distance ranging from 0.2 to 3.8 mm (Fig. 9.4).[114] Another group of authors, using CBCT, reported that the most frequent position of the apical foramen was central but it varied.[90] Root resorption is an additional factor in length determination. Resorption is more common with pulp necrosis and apical bone resorption and this can result in loss of the constriction (Fig. 9.5).[94,173] On

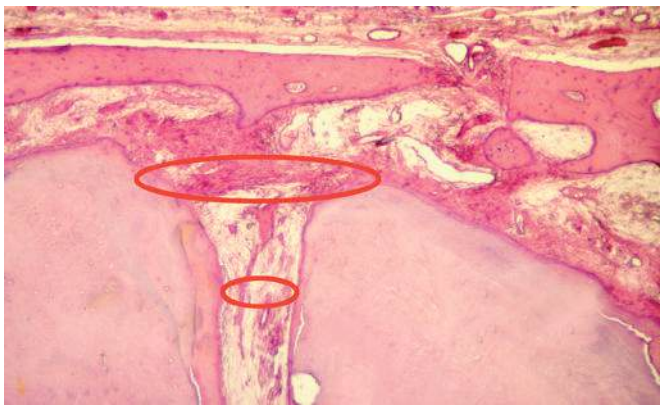


Fig. 9.3 Histologic section of a root apex, demonstrating anatomy of the classic foramen and constriction.

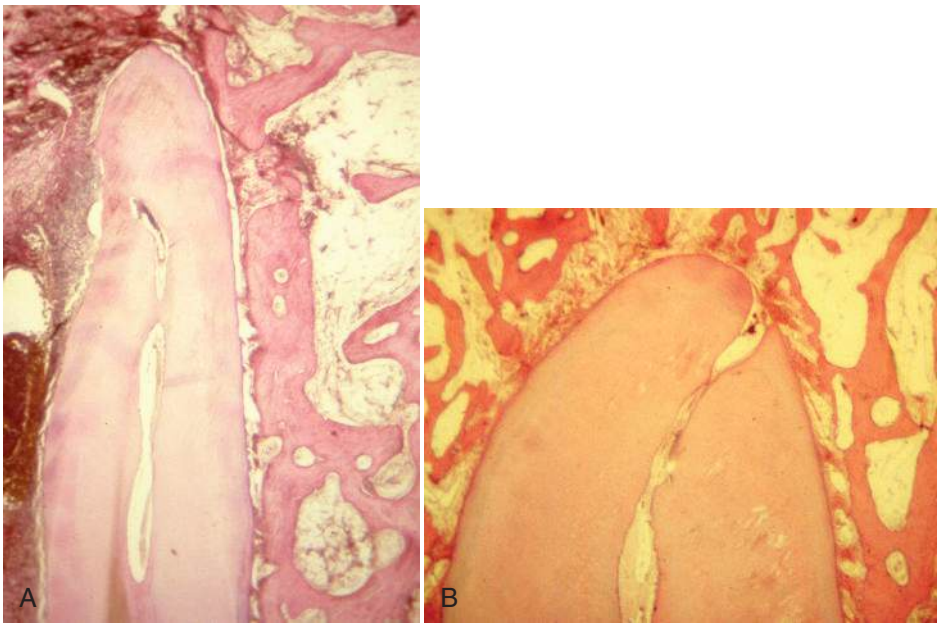


Fig. 9.4 Histologic sections demonstrating the foramen exiting short of the root apex.



Fig. 9.5 Scanning electron microscopy of a tooth exhibiting a necrotic pulp and apical pathosis and resorption.

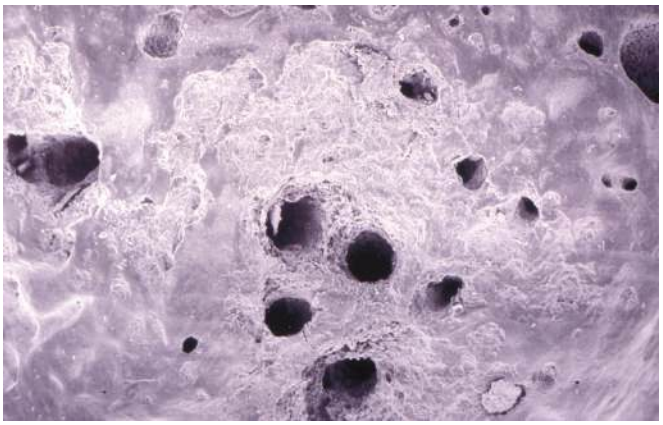


Fig. 9.6 Scanning electron microscopy of the apex of an extracted tooth that was removed because of pulp necrosis. Note the multiple accessory foramina and resorption.

the basis of these findings, it appears that an RCS obturated to the radiographic apex reflect an overextension of the obturating material. If overextension occurs that cannot be retrieved, and there appears to be resultant nerve damage, the practitioner is obligated to refer the patient to a qualified practitioner skilled in cases of this type.[105]

In a study by the Toronto group on the prognosis of retreatment identified perforation,[73] pretreatment periradicular disease and adequate length of the root canal filling were identified as factors significantly influencing healing or nonhealing outcomes. The authors speculated that canals obturated more than 2 mm short of the radiographic apex harbored necrotic tissue, bacteria, and irritants that, when retreated, could be cleaned and sealed. The success rate for negotiating the apical unfilled canal was 74%.

Controversy also exists regarding the role that accessory canals play in healing and nonhealing outcomes (Fig. 9.6). A scanning electron microscopy (SEM) study of the apical anatomy of each tooth group except third molars noted no pattern for foraminal openings.[113] The number of accessory canals ranged from 1 to 16. Although lateral canals can be associated with pathosis, one study that examined root-treated teeth from human cadavers reported no relationship between unfilled lateral canals and periradicular pathosis.[21] Accessory/lateral canals are often obturated by chance and only serendipitously identified on a posttreatment radiograph (Fig. 9.7).

A subsequent histobacteriological study also found no evidence to support the concept that lateral canals must be obturated to achieve a healing long-term treatment outcome (Fig. 9.8).[236] In cases with vital pulps, forcing obturation materials into lateral canals resulted in unnecessary damage to the periradicular tissues with consequential inflammation. In cases with necrotic, infected pulps in which lateral canals appeared radiographically obturated, they were actually not sealed or disinfected. Moreover, the remaining tissue in the ramification was inflamed and enmeshed with obturation materials. This, however, does not

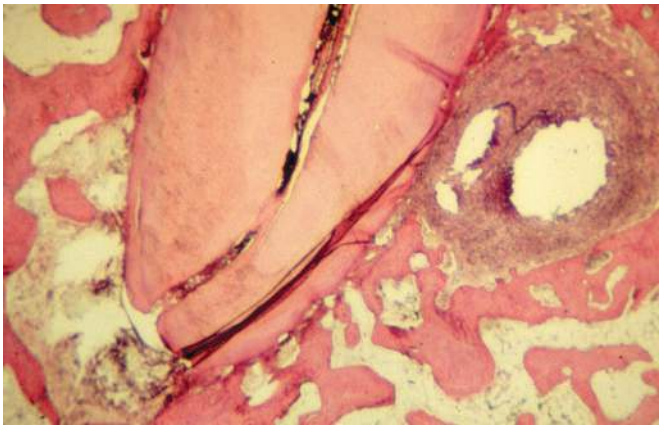


Fig. 9.7 Histologic section of a mesial root of a mandibular molar with a lateral canal present and associated lesion. Will the lesion resolve after the removal of the main canal contents, or will the lesion persist because of necrotic pulpal remnants in the lateral canal? The question remains unanswered.



Fig. 9.8 Posttreatment radiograph of a mandibular right first molar with a lateral canal associated with the distal root.

mean that lateral canals should not be optimally debrided and disinfected by contemporary irrigant delivery and agitation techniques to reduce microbes and/or biofilms that reside within those spaces prior to obturation of the RCS with obturation materials and sealers.

The importance of length control in obturation relates to possible overextension of root filling materials. In a quantitative systematic review of the literature,[255] the authors critically analyzed 12 outcome studies that fulfilled the following criteria: (i) a minimum follow-up period of 2 years; (ii) data available on termination of obturation/instrumentation; (iii) adequate definition of treatment failure; (iv) data available on healing/nonhealing outcomes of NS RCT in relation to the obturation/instrumentation length; and (v) presence or absence of a periapical radiolucency. The 12 studies were classified into three categories based on the obturation length from the radiographic apex: (A) 0 to 1 mm; (B) >1 mm but <3 mm; and (C) obturated past the radiographic apex, including sealer. From the 12 studies, only 4 studies that included data that could be placed into the three length categories were further subjected to meta-analysis. Results of the meta-analysis indicated that the healing rate in group A (obturated 0 to 1 mm from apex) was marginally better than group C (obturated past apex) by 29%. Although group A demonstrated better healing than group B (obturated >1 mm short), the difference was not statistically significant. The authors concluded that a higher healing rate was achieved when treatment involved obturation short of the apex.

The aforementioned results derived from the sophisticated meta-analysis of multiple publications highlighted the data presented in the classic work performed by Sjögren and coworkers on healing of root-treated teeth with necrotic pulps and PL 8 to 10 years after treatment.[275] When those teeth were obturated within 2 mm of the apex, 94% revealed normal periapical conditions with the use of periapical radiography at the follow-up examination. Conversely, roots with overextended root fillings and those with obturations more than 2 mm short of the apex demonstrated significantly lower healing rates of 76% and 68%, respectively.

Similar results were also reported in a prospective clinical study on the healing[194] and survival[194] of teeth that had undergone primary or secondary NS RCT for 2 to 4 years. Absence of root-filling overextension was found to be a highly significant postoperative prognostic factor affecting the healing of both primary and secondary NS RCT.[194] With respect to tooth survival, overextension of GP obturations did not have any effect on tooth survival within the first 22 months but significantly increased the risk of tooth loss beyond 22 months.[194] This prognostic factor was common to both primary and secondary NS RCT. The authors attributed the delayed effect of overextension of GP obturations on tooth survival to the possibility of subcritical cracks created by excessive forces during compaction of GP, which eventually propagated during function and ultimately resulted in catastrophic failure.

On the basis of biologic and clinical principles, the conclusions derived from studies performed for more than two decades apart are all agreeable in that instrumentation and obturation should not extend beyond the apical foramen.[191] This was also demonstrated in a histological study that evaluated 41 human root-filled teeth from 36 patients.[234] In six cases exhibiting overextensions, histologic examination revealed severe inflammation.

Whereas the guideline of 1 mm from the radiographic apex remains rational when using radiographs, the point of apical termination of the preparation and obturation remains empirical. The use of an apex locator in conjunction with radiographs and sound clinical judgment makes this decision more logical. The need to compact the GP and sealer against the apical dentin matrix is necessary to prevent overextension of obturation materials into the periapical tissues. Deciding where the apical constriction of the RCS lies is based on the clinician's basic knowledge of apical anatomy, tactile sensation, radiographic interpretation, apex locators, apical bleeding, and, if not anesthetized, the patient's response.

Preparation for Obturation

During the C&S process, organic pulpal materials and inorganic dentinal debris accumulate on the canal wall, producing an amorphous irregular smear layer (Fig. 9.9).[170,178,215]

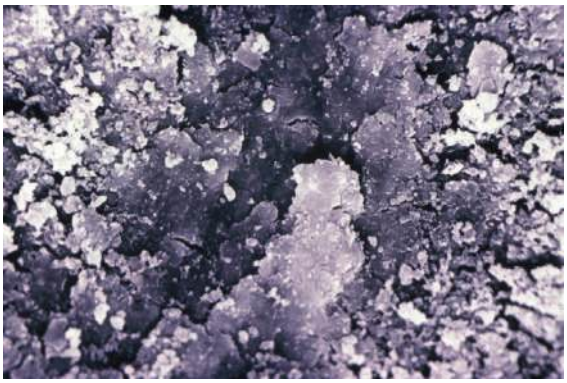


Fig. 9.9 Scanning electron microscopy of a prepared canal wall. The tubules are covered with a smear layer of organic and inorganic material.

As shown in one of these studies, the smear layer is superficial, with a thickness of 1 to 5 μm.[178] This superficial debris can be packed into the dentinal tubules to various distances.[3]

In cases of pulpal necrosis, this layer may also be contaminated with bacteria and their by-products. For example, one study reported that bacteria can extend 10 to 150 μm into the dentinal tubules of necrotic teeth.[261] Another study noted that capillary action and fluid dynamics play a role in packing debris into the tubules.[5] Another investigation noted a mean penetration of 479 μm after a 28-day incubation period.[219]

The smear layer is not a complete barrier to bacteria but may act as a physical barrier, decreasing bacterial penetration into tubules. This was illustrated by a study reporting that removal of the smear layer permitted colonization of the dentinal tubules at a significantly higher rate when compared with leaving the smear layer in place.[82]

The smear layer may also interfere with adhesion and penetration of sealers into dentinal tubules.[305] Evidence indicates that sealer penetration into dentinal tubules does not occur when the smear layer is present.[99,211] For example, one study reported that removal of the smear layer permitted Roth 811 (Roth International, Ltd., Chicago, IL, USA), Calciobiotic root canal sealer (CRCS; Coltène/Whaledent, Cuyahoga Falls, OH, USA), and Sealapex (SybronEndo, Orange, CA, USA) to penetrate to between 35 and 80 μm, whereas the presence of the smear layer obstructed tubular penetration of all sealers.[297] Other studies reported that smear layer removal increased bond strength and reduced microleakage in teeth obturated with AH-26 (DENTSPLY Maillefer, Ballaigues, Switzerland).[86,100] Another investigation reported that a combination of smear layer removal, AH-26 as the sealer, and vertical compaction of GP had a cumulative effect in reducing leakage.[289]

There does not appear to be a consensus on removing the smear layer before obturation and the advantages/disadvantages of smear layer removal remains inconclusive.[59,262,265] However, growing evidence supports removal of the smear layer before obturation.[131,194,265] The organic debris present in the smear layer might constitute a substrate for bacterial growth.[215] It has been suggested that the smear layer prohibits sealer contact with the canal wall and permits leakage.[27] Bacterial penetration in the presence of a smear layer in RCS obturated with thermoplasticized GP and sealer has been shown to be significantly higher than with smear layer removal before obturation.[215] An additional consideration is the presence of viable bacteria that remain in the dentinal tubules and use the smear layer for sustained growth and activity.[39] Removal of the smear layer introduces the possibility of reinfecting the dentinal tubules if leakage occurs.[262] However, one study demonstrated that bacteria present before obturation are not viable after obturation.[77]

The smear layer may also interfere with the action of irrigants used as disinfectants.[204] When the smear layer is not removed, the dentinal tubules provide a long-term sanctuary for previously entrapped microorganisms to participate in reinfections. Additionally, the smear layer may slowly disintegrate and dissolve around leaking obturation materials or it may be removed by bacterial by-products such as acids and enzymes.[262]

The smear layer may interfere with the adhesion and penetration of root canal sealers. It also may prevent GP penetration during thermoplastic obturation techniques.[115] Significant tubular penetrations of GP and sealers have been reported with thermoplasticized obturations[115] and with dentin-bonded composite resins.[162] Removal of the smear layer also enhances the adhesion of sealers to dentin and tubular penetration.[162,201,262,304] Root canal filling materials adapt better to the canal walls after smear layer removal.[79,201,304,305]

An additional method for removing the smear layer involves sonic and ultrasonic instruments. In early studies of ultrasonic instrumentation, investigators noted the technique was effective in removing the smear layer.[69] Another investigator also demonstrated smear layer removal with ultrasonication and NaOCl.[53] One study compared the cleaning efficacy of short-term sonic and ultrasonic passive irrigation with 5.25% NaOCl after hand instrumentation in the apical 3 to 6 mm of maxillary molar root canals.[241] Passive sonic or ultrasonic irrigation for 30 seconds resulted in significantly cleaner canals than hand filing alone, and ultrasonic irrigation produced significantly cleaner canals than irrigation. However, other studies found ultrasonication and NaOCl to be ineffective in removing the smear layer.[23,300] In a more recent study, a group of investigators reported that the use of the Vibringe, EndoActivator, or needle irrigation did not significantly improve sealer penetration when compared with conventional irrigation.[36]

After the completion of C&S procedures, removal of the smear layer is generally accomplished by irrigating the canal with 17% disodium EDTA and 5.25% NaOCl (Fig. 9.10).[24] Chelators remove the inorganic components leaving the organic tissue elements intact. Sodium hypochlorite is necessary for removal of the remaining organic components. Citric acid has also been shown to be an effective method for removing the smear layer,[22,125] as has doxycycline[20] and tetracycline.[121]

Chelating agents were introduced to endodontic treatment by Nygaard-Østby in 1957 for treatment of calcified narrow RCS.[199] EDTA is the chelating solution customarily used in endodontic treatment. It is available in both liquid and viscous forms with common concentrations between 15% and 17%,[131] and its effectiveness is related to time, pH, and concentration.[199] Demineralization results in increased dentin permeability[112] because of the removal of the smear layer and plugs and enlargement of the tubules. It appears that the tubular enlargement is due to selective removal of



Fig. 9.10 Scanning electron microscopy of the canal wall after removal of the smear layer with 17% EDTA and 5.25% sodium hypochlorite.

the peritubular dentin.[129] The action of chelators and acids appears to be more effective in the coronal and middle thirds of the root and is reduced apically.[131,167] This reduced activity may reflect canal size.[210] This is a clinical concern because of the more irregular structure of dentin in the apical third. Another investigation demonstrated marked variations in the apical portion of the root,[183] including accessory RCS, areas of resorption and repaired resorptions, pulp stones, irregular or absent primary tubules, irregular secondary dentin, and cementum-like tissue lining the apical root canal wall. The variable structure of the apical region of human teeth presents challenges to the use of endodontic obturation techniques requiring adhesives, because this may influence the dentin bonding ability in the apical region.[183]

Although EDTA is mildly cytotoxic,[299] the irrigant appears to be biocompatible when used clinically;[199] however, irreversible decalcification of periapical bone and neuroimmunological disturbances have been noted.[259] EDTA decreases the microhardness of dentin.[67] Extrusion of both NaOCl and EDTA in clinical treatment should be avoided.[120,216,286]

The recommended time for removal of the smear layer is 1 to 5 minutes.[51,131,254] The small particles of the smear layer are primarily inorganic with a high surface-to-mass ratio that facilitates removal by acids and chelators. Investigators have found that a 1-minute exposure to 10 mL of EDTA was adequate to remove the smear layer and that a 10-minute exposure caused excessive removal of both peritubular and intratubular dentin.[51]

The use of EDTA in combination with NaOCl is recommended[24,316] and may enhance the cleaning[167] and antimicrobial effects of these solutions when compared with using them alone.[49] Although the use of EDTA alone does not cause erosion of the canal wall dentin, it should be noted the adjunctive use of NaOCl with EDTA may result in erosion of the intraradicular dentin depending on the timing and the concentrations of irrigants employed.[171,227,319]

New smear layer–removing irrigants are commercially available that combine calcium chelation of the inorganic component of the smear layer with antimicrobial activities. Examples of these irrigants include QMix 2in1 Irrigating Solution (DENSPLY Tulsa Dental Specialties).[16,70]

The Ideal Root Canal Obturation Materials

Various endodontic materials have been advocated for obturation of the RCS. Most techniques employ a core material and sealer. Regardless of the core material, a sealer is essential to every technique and helps achieve a fluid-tight seal.

The American Association of Endodontists' *Guide to Clinical Endodontics* outlines contemporary endodontic treatment.[2] NS RCT of permanent teeth "involves the use of biologically acceptable chemical and mechanical treatment of the RCS to promote healing and repair of the periradicular tissues." The process is accomplished under aseptic conditions with rubber dam isolation. Regarding obturation, the guide states, "Root canal sealers are used in conjunction with a biologically acceptable semi-solid or solid obturating material to establish an adequate seal of the RCS." In this area, the guidelines indicate, "Paraformaldehyde-containing paste or obturating materials have been shown to be unsafe. Root canal obturation with paraformaldehyde-containing materials is below the standard of care for endodontic treatment" (Fig. 9.11). Chapter 27 gives further information about this issue.

Assessment of nonsurgical treatment is based primarily on the posttreatment radiographic examination. The radiographic criteria for evaluating obturation include the following categories: length, taper, density, GP and sealer removal to the facial cementoenamel junction in anterior teeth and to the canal orifice in posterior teeth, and an adequate provisional or definitive restoration (Fig. 9.12).

Quality assurance is accomplished through a careful evaluation of treatment procedures. Only by this approach can deficiencies be identified and corrected. Although the



Fig. 9.11 A periapical radiograph of a mandibular left second premolar and first molar, demonstrating Sargenti paste NS RCT. In addition to the toxic material, the technique often accompanies inadequate C&S procedures.

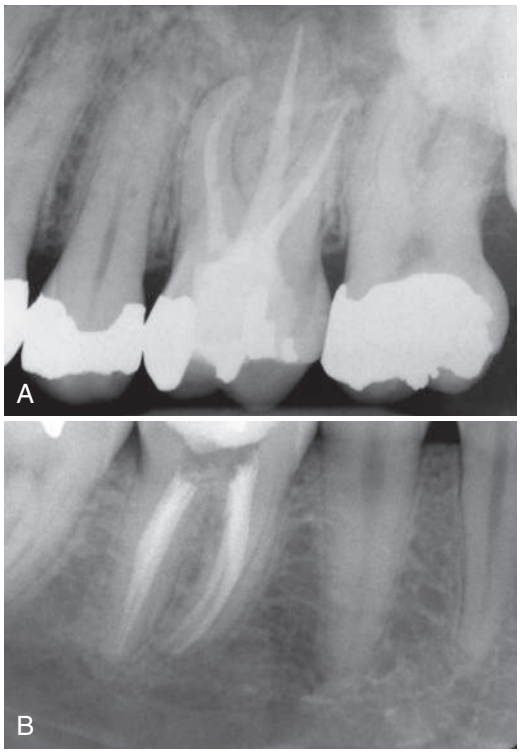


Fig. 9.12 **A,** Posttreatment radiograph of a maxillary right first molar demonstrating adequate length, density, and taper. **B,** Posttreatment radiograph of a mandibular right first molar with an adequate obturation.

anatomy and morphology of RCS vary tremendously, the obturated RCS should reflect the original shape of the RCS. Procedural errors in preparation, such as loss of length control, ledging, apical transportation, apical perforation, stripping perforation, and separated instruments, may not be correctable. Errors in obturation, such as length, voids, inadequate removal of prior obturation materials, and temporization, may be correctable.

Radiographic interpretation may vary among clinicians because of differences in radiopacity in root canal sealer/cements, constituents in specific brands of GP, interpretation of voids in vivo versus in vitro,[318] the overlying bony anatomy, radiographic angulation, and the limited two-dimensional view of the obturated RCS.

An often overlooked aspect in the assessment of RCS obturation is the density of the apical portion of the obturation. The apical third of the RCS may be filled with a sea of root canal cement and a single master cone or poorly compacted mass of previously softened GP. Radiographically, the apical third of the canal appears less radiodense. An ill-defined outline to the canal wall is evident, along with obvious gaps or voids in the filling material or its adaptation to the confines of the RCS. Because of the use of highly radiopaque root canal sealers/cements, the apical portion may be filled only with sealer giving the clinician the false impression of a dense, three-dimensional obturation with GP.

Root canal sealers vary in radiopacity.[230,285] Some contain silver particles or significant amounts of barium sulfate to enhance their radiopacity. Although these components may enhance the visualization of anatomic structures such as lateral canals, it is important to realize they do not increase the sealing ability of the sealer and the quality of the obturation. They may also give the impression that an RCS is well obturated when voids are masked by the density of the sealer. It is erroneous to claim that obturations with highly radiopaque sealers are better than those made with less radiopaque materials. This type of comparison and claim to superiority are both unfounded and unwarranted. The radiographic appearance or aesthetic appearance of the obturated canal system should be secondary to meticulous C&S. Although assessment of the RCS obturation is based on radiographic findings, root canal sealers do not have to be highly radiopaque to be effective.

Types of Sealers

Root canal sealers are necessary to seal the space between the dentinal wall and the obturating core interface. Sealers also fill voids and irregularities in the RCS, lateral and accessory canals, and spaces between GP points used in lateral compaction. Sealers also serve as lubricants during the obturation process. Grossman outlined the properties of an ideal sealer (Box 9.1).[110] Presently, no sealer satisfies all the criteria.

Sealers should be biocompatible and well tolerated by the periradicular tissues.[140] All sealers exhibit toxicity when freshly mixed; however, their toxicity is greatly reduced on setting.[142,155] Sealers are resorbable when exposed to tissues and tissue fluids, but not all extruded sealers are predictably removed from the periradicular tissues.[17,235] Tissue healing and repair generally appear unaffected by most sealers, provided there are no adverse breakdown products of the sealer over time.[35,40-42,44] Breakdown products from sealers may have an adverse effect on the proliferative capability of periradicular cell populations, specifically in the presence of the apical lesion.[235] As a result, sealers should not be placed routinely in the periradicular tissues as part of an obturation technique.

The most popular sealers are zinc oxide–eugenol formulations, $Ca(OH)_2$ sealers, glass ionomer sealers, resin-based epoxy resin or methacrylate resin sealers, and the recently introduced calcium silicate–based sealers. Despite claims by the manufacturers on the advantages of each class, there is no evidence-based data, based on randomized clinical trials, demonstrating the superiority of one sealer over another. Regardless of the sealer selected, all exhibit toxicity until they have set. For this reason, extrusion of sealers into the periradicular tissues should be avoided (Fig. 9.13).[246]

Box 9.1 Properties of an Ideal Sealer

- Exhibits tackiness when mixed to provide good adhesion between it and the canal wall when set
- Establishes a hermetic seal
- Radiopaque, so that it can be seen on a radiograph
- Very fine powder, so that it can mix easily with liquid
- No shrinkage on setting
- No staining of tooth structure
- Bacteriostatic, or at least does not encourage bacterial growth
- Exhibits a slow set
- Insoluble in tissue fluids
- Tissue tolerant; that is, nonirritating to periradicular tissue
- Soluble in a common solvent if it is necessary to remove the root canal filling

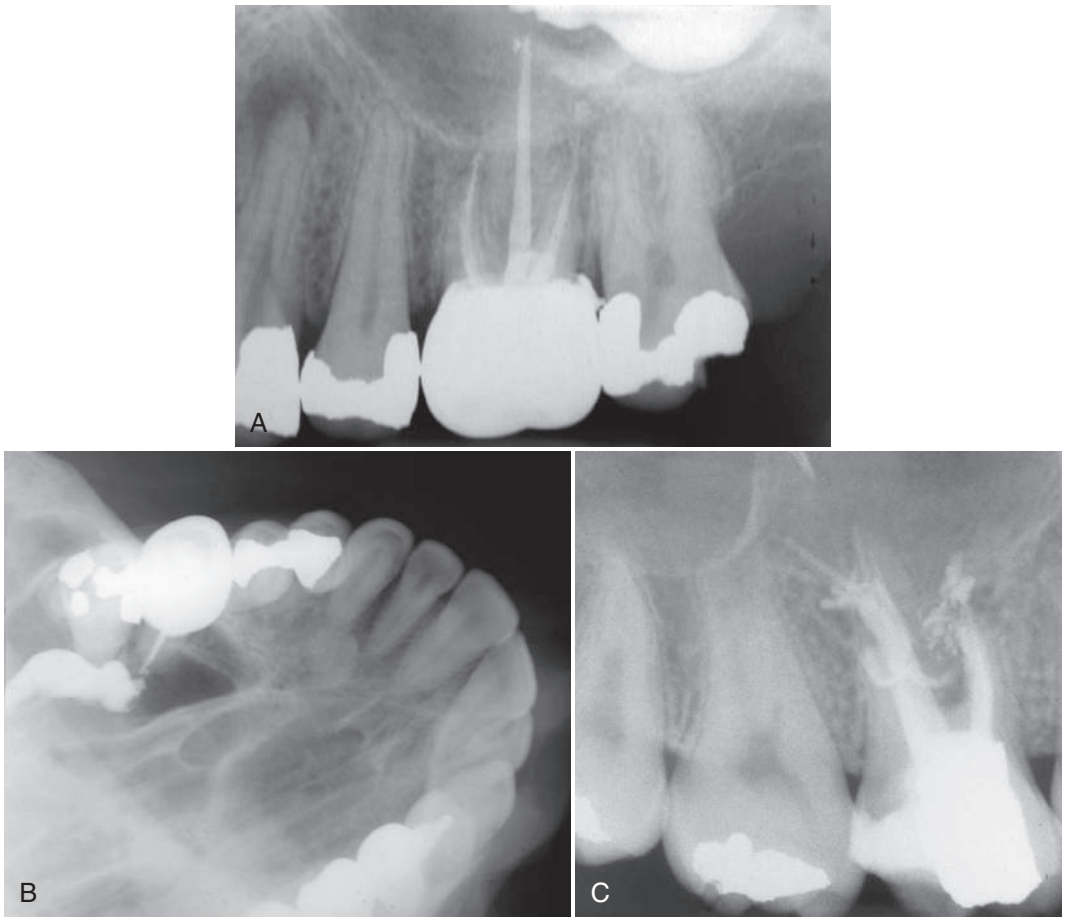


Fig. 9.13 **A,** Extrusion of sealer evident on the posttreatment radiograph of a maxillary first molar. The separated lentulo spiral in the mesiobuccal root indicates a possible method of sealer placement. **B,** Maxillary occlusal film demonstrates that the sealer is located in the maxillary sinus. Correction by nonsurgical techniques is not possible. **C,** Maxillary right first molar with extrusion of the sealer and GP.

ZINC OXIDE AND EUGENOL

Zinc oxide–eugenol sealers have a history of successful use over an extended period of time. Zinc oxide–eugenol sealers will resorb if extruded into the periradicular tissues.[17] They exhibit a slow setting time,[7] shrinkage on setting,[143] solubility,[221] and they can stain tooth structure.[151] An advantage to this sealer group is antimicrobial activity.[6,19,123]

Marketed as Pulp Canal Sealer (Kerr Endo) and Pulp Canal Sealer EWT (extended working time), this sealer is popular with clinicians using thermoplastic techniques.

Grossman modified the formulation and introduced a nonstaining formula in 1958 but it has been discontinued (Table 9.1),[111] but Pulp Canal Sealer EWT (>6 hours on the mixing pad) and EWT (Kerr) have a similar formulation. Tubli-Seal (Kerr) is a catalyst/base zinc oxide–eugenol sealer that is convenient to mix but has a faster setting time when compared with the liquid/powder sealers. Tubli-Seal EWT provides an extended working time.

Although zinc oxide–eugenol sealers possess marked cytotoxic and tissue-irrigating potencies in ex vivo cell culture studies,[142] their clinical usefulness has been well demonstrated in retrospective and prospective human clinical studies.[66,291] This discrepancy between the results of in vitro and in vivo cytotoxicity testing may be explained by the fact that most cell culture systems are represented by only one cell type, that is, no cell-cell interactions, which is often monoclonal in origin. Another important issue to consider is the fact that culture conditions are not homeostatic and there is no elimination of toxic substances as there would be in vivo. By contrast, the human body possesses a lymphatic system and periapical defenses such as polymorphonuclear leukocytes, plasma cells, and macrophages to help eliminate toxic substances.[55] These mechanisms do not exist in a culture plate and must be taken into account for interpretations of the results of cell culture-based cytotoxicity studies reported in the endodontic literature.

Table 9.1 Formula for Zinc Oxide–Eugenol Root Canal Sealer

Powder	Liquid
Zinc oxide	42 parts
Staybelite resin	27 parts
Bismuth subcarbonate	15 parts
Barium sulfate	15 parts
Sodium borate, anhydrous	1 part

CALCIUM HYDROXIDE SEALERS

Calcium hydroxide sealers were developed for therapeutic activity. It was thought that these sealers would exhibit antimicrobial activity and have osteogenic–cementogenic potential. Unfortunately, these actions have not been demonstrated.[78,184] Solubility is required for release of $Ca(OH)_2$ and sustained activity. This is inconsistent with the purpose of a sealer. CRCS is a zinc oxide–eugenol sealer with $Ca(OH)_2$ as one ingredient. Sealapex (SybronEndo) is a catalyst/base system. The base contains zinc oxide, $Ca(OH)_2$, butyl benzene, sulfonamide, and zinc stearate. The catalyst contains barium sulfate and titanium dioxide as radiopacifiers in addition to resin, isobutyl salicylate, and aerosol R972. Apexit and Apexit Plus (Ivoclar Vivadent, Schaan, Liechtenstein) consist of an activator (disalicylate, bismuth hydroxide/bismuth carbonate, and fillers) and a base [$Ca(OH)_2$, hydrated colophonium (i.e., pine resin) and fillers].

Although an osteogenic response has been observed with $Ca(OH)_2$-based root canal sealers,[128,276,284] the ability of these sealers to sustain a high pH over time and produce an alkaline pH at the root surface has been questioned.[89]

Better biocompatibility and osteogenic potential have been reported by $Ca(OH)_2$-producing tricalcium silicate-based root canal sealers than those with conventional $Ca(OH)_2$ sealers (Sealapex).[61]

NONEUGENOL SEALERS

Developed from a periodontal dressing, Nogenol (GC America, Alsip, IL) is a root canal sealer without the irritating effects of eugenol. The base contains zinc oxide, barium sulfate, and bismuth oxychloride.

GLASS IONOMER SEALERS

Glass ionomers have been advocated for use in obturation because of their dentin-bonding properties. Ketac-Endo (3M ESPE, Minneapolis, MN, USA) enables adhesion between the material and the canal walls.[98] It has been reported to have compatible outcomes compared with previous studies.[98] However, it is difficult to properly treat the dentinal walls in the apical and middle thirds with conditioning agents to receive the glass ionomer sealer. A disadvantage of glass ionomers is that they are more difficult to remove if retreatment is required.[187] This sealer has minimal antimicrobial activity.[63]

Activ GP (Brasseler USA) consists of a glass ionomer–impregnated GP cone with a glass ionomer external coating and a glass ionomer sealer (Fig. 9.14). Available in 0.04 and 0.06 tapered cones, the sizes are laser verified to ensure a more precise fit. This single cone technique is designed to provide a bond between the dentinal canal wall and the master GP cone that would be impregnated with glass ionomer filler particles resulting in an alleged monoblock.[95]

RESIN SEALERS

Resin sealers have a long history of use, provide adhesion, and do not contain eugenol. There are two major categories: epoxy resin and methacrylate resin.



Fig. 9.14 Activ GP (Brasseler USA, Savannah, GA) glass ionomer–coated GP points and sealer.

Epoxy Resin Sealers

AH-26 (DENTSPLY DeTrey, Konstanz, Germany) is a slow-setting epoxy resin that was found to release formaldehyde when setting.[149,278] AH Plus (DENTSPLY DeTrey) is a modified formulation of AH-26 in which formaldehyde is not released (Fig. 9.15).[163] The sealing abilities of AH-26 and AH Plus appear comparable. AH Plus is an epoxy resin-amine based system that comes in two tubes. The epoxide paste tube contains a diepoxide (bisphenol A diglycidyl ether) and fillers as the major ingredients, while the amine paste tube contains a primary monoamine, a secondary diamine, a disecondary diamine, silicone oil, and fillers as the major ingredients. It exhibits a working time of approximately 4 hours.

Methacrylate Resin Sealers

Four generations of methacrylate resin-based root canal sealers have been marketed for commercial use.[147,212] The first generation of hydrophilic methacrylate resin-based material (Hydron; Hydron Technologies, Inc., Pompano Beach, FL, USA) was designed for en masse obturations and appeared in the mid-1970s when scientific foundations



Fig. 9.15 AH Plus sealer is a resin formulation. (Courtesy DENTSPLY, Konstanz, Germany. All rights owned by and used with permission from DENTSPLY International, Inc.)

behind dentin bonding were in their infancy stage of development. The major component of Hydron was poly [2-hydroxyethyl methacrylate] (poly [HEMA]), which was injected into an RCS and polymerized in situ within the RCS without the adjunctive use of a core obturation material. Hydron became obsolete in the 1980s as subsequent clinical outcomes were unacceptable.[156]

Following the marketing of self-priming, self-etching, and self-adhesive resin cement technologies in restorative dentistry, functionally analogous, low viscosity methacrylate resin-based root canal sealers have since been available for use in endodontics. This type of bondable root canal sealers has been aggressively promoted with the highly desirable property of creating monoblocks within the RCS space.[288] The term *monoblock* refers to the idealized scenario in which the canal space becomes perfectly filled with a gap-free, solid mass that consists of different materials and interfaces, with the purported advantages of simultaneously improving the seal and fracture resistance of an obturated RCS.

The second generation of bondable sealer is nonetching and hydrophilic in nature and does not require the adjunctive use of a dentin adhesive. It is designed to flow into accessory canals and dentinal tubules to facilitate resin tag formation for retention and seal after smear layer removal with NaOCl and EDTA. EndoREZ (Ultradent Products Inc., South Jordan, UT, USA) is a dual-cured radiopaque hydrophilic methacrylate sealer that contains nonacidic diurethane dimethacrylate. The addition of triethyleneglycol dimethacrylate to the sealer composition renders it hydrophilic, so that it may be used in the wet environment of the RCS and be very effective in penetrating dentinal tubules and forming long resin tags.[29,287] The sealer was found to seal best when applied to slightly moist intraradicular dentin.[324] EndoREZ is recommended for use with either a conventional GP cone or specific EndoREZ points that are resin-coated GP. A retrospective clinical and radiographic study evaluating the 10-year treatment outcome of one-visit NS RCT using GP and the EndoREZ sealer reported accumulative probability of success of 92.1% after 10 years.[323] The authors concluded that EndoREZ may be recommended as an alternative to other commonly used root canal sealers. Unfortunately, no additional root canal sealer was used in that study for comparison.

New generations of self-etching (third generation) and self-adhesive (fourth generation) resin cements have been introduced to restorative dentists to simplify bonding procedures. They became commercially available shortly after the introduction of those resin cement systems. The third-generation self-etching sealers contain a self-etching primer and a dual-cured resin composite root canal sealer. The use of self-etching primers reintroduced the concept of incorporating smear layers created by hand/rotary instruments along the sealer-dentin interface. An acidic primer is applied to the dentin surface that penetrates through the smear layer and demineralizes the superficial dentin. The acidic primer is air-dried to remove the volatile carrier and then a dual-cured moderately filled flowable resin composite sealer is applied and polymerized. Provided that these materials are sufficiently aggressive to etch through thick smear layers, the technique sensitivity of bonding to RCS may be reduced when smear layers are inadvertently retained in the apical third of instrumented canal walls.

Third-generation methacrylate resin-based sealers that incorporate the use of self-etching primers became popularized following the introduction of Resilon (Resilon Research LLC, Madison, CT, USA), a dimethacrylate-containing polycaprolactone-based thermoplastic root filling material.[266] Resilon has been removed from the market and is no longer available.

The fourth-generation methacrylate resin-based sealers (e.g., MetaSEAL, Parkell Inc., RealSeal SE, SybronEndo) is functionally analogous to a similar class of recently introduced self-adhesive resin luting cements in that they have further eliminated the separate etching/bonding step.[229] Acidic resin monomers that are originally present in dentin adhesive primers are now incorporated into the resin-based sealer/cement to render them self-adhesive to dentin substrates. The combination of an etchant, a primer, and a sealer into an all-in-one self-etching, self-adhesive sealer is advantageous in that it reduces the application time as well as errors that may occur during each bonding step. MetaSEAL is the first commercially available fourth-generation self-adhesive dual-curable sealer.[158] The sealer purportedly bonds to thermoplastic root-filling materials as well as radicular dentin via the creation of hybrid layers in both substrates. MetaSEAL is also marketed as Hybrid Bond SEAL (Sun Medical Co. Ltd., Shiga, Japan) and had been reported to produce equivalent or slightly inferior sealing properties as conventional nonbonding epoxy resin-based sealers.[28,202]

SILICONE SEALERS

RoekoSeal (Coltène/Whaledent AG, Altstätten, Switzerland) is a polydimethylsiloxane that has been reported to expand slightly on setting.[205]

GuttaFlow and GuttaFlow2 (Coltène/Whaledent) are cold flowable matrices that are triturated. They consist of GP in a particulate form less than 30 μm added to RoekoSeal (Fig. 9.16). The material is provided in capsules for trituration. The technique involves injection of the material into the RCS, followed by placement of a single master cone. The material provides a working time of 15 minutes and it cures in 25 to 30 minutes. Evidence suggests that the material fills canal irregularities with consistency[320] and is biocompatible,[65,174] but the setting time is inconsistent and may be delayed by final irrigation with sodium hypochlorite.[37]



Fig. 9.16 GuttaFlow trituration capsule and injection syringe (Coltène/Whaledent, Cuyahoga Falls, OH).

GuttaFlow bioseal, the most recent introduction to this family of silicone sealers, incorporates bioactive ingredients such as calcium and silicate into the original formulation, which is claimed by the manufacturer to stimulate tissue healing and regeneration.[52] A recent study reported that GuttaFlow bioseal promotes spontaneous differentiation of human periodontal ligament stem cells into cementoblast-like cells in the absence of adjunctive growth factors.[238] If these results can be validated in in vitro experiments, GuttaFlow bioseal should have the potential to promote the regeneration of the cementum in teeth with apical periodontitis or chronic abscess.

TRI/DICALCIUM SILICATE SEALERS

Root canal sealers based on tricalcium silicates and dicalcium silicates are the latest group of bioactive root canal sealers introduced for obturation of an RCS.[135] These sealers are an outgrowth of the popularity of mineral trioxide aggregate (MTA) materials, which are based on tricalcium silicate and dicalcium silicate, a water-setting hydraulic powder used for various surgical and vital pulp therapy treatments.[18,214,292] This type of root canal sealer is attractive because of the bioactivity that has been reported for MTA-type materials,[198] which are also known for being hydrophilic.[54]

MTA is a hydraulically active powder that contains primarily tricalcium silicate, dicalcium silicate, and a radiopaque powder, often bismuth oxide. Because of staining of crown dentin by the bismuth oxide component, which may be rendered brown in the presence of NaOCl, gray in the presence of chlorhexidine, or even black in the presence of glutaraldehyde, the radiopacifier has now been replaced with other materials such as zirconia dioxide or tantalum oxide in some commercial formulations. The radiopaque agent, bismuth oxide, zirconia, or tantalum oxide, is important, without which MTA would not be distinguishable on a radiograph. Minor phases of tricalcium aluminate and calcium sulfate are often present. Some MTA-type products contain calcium carbonate, or tetracalcium aluminoferrite. The term MTA is used here to denote all the tri- and dicalcium silicate products.

Tri/dicalcium silicate cements/sealers are set by reaction with water and form a highly alkaline (pH ~ 12) mixture consisting of a rigid matrix of calcium silicate hydrates and $Ca(OH)_2$.[68,71] These hydrates form on the surface of the original calcium silicate particles, and hydration gradually penetrates inward. When a tricalcium silicate cement sets, the dimensional change is less than 0.1% expansion, which helps with creating a barrier, which is especially important for its use during endodontic treatment. Setting time for tricalcium silicate cements is lengthy, about 165 minutes for the initial set and less than 6 hours for the final set, which has been a major drawback for using them in some indications, but not for its use as a sealer. During the early stage of development, the inconsistency of MTA products with water has rendered it not suitable for use as a sealer. Additionally, the coarseness of the first MTA products, intended for surgical uses, made them unsuitable for use as a sealer, as their film thickness was too high (>50 μm).

To date, many tri/dicalcium silicate-based sealers have been available commercially (Figs. 9.17–9.19).[135,166,179,267,296,301]



Fig. 9.17 BioRoot RCS (Septodont, Saint-Maur-des- Fossés, France).



Fig. 9.18 iRoot SP (Innovative BioCeramix Inc., Vancouver, Canada; *aka* EndoSequence BC sealer; Brasseler USA, Savannah, GA).



Fig. 9.19 MTA Plus (NuSmile, Houston, TX).

They include the following: MTA Fillapex (Ângelus Indústria de Produtos Odontlógicos Ltda; Londrina, Paraná, Brazil), iRoot SP (Innovative BioCeramix Inc., Vancouver, Canada; *aka* Endosequence BC sealer; Brasseler USA, Savannah, GA, USA), Endo CPM Sealer (EGEO SRL, Buenos Aires, Argentina), MTA Plus (NuSmile, Houston, TX, USA), Endoseal MTA (Maruchi, Wonju, Korea), Total Fill BC Sealer (FKG Dentaire, La-Chaux-de-Fonds, Switzerland), BioRoot RCS (Septodont, Saint-Maur-des- Fossés, France), Seal Plus BC (MK Life, Porto Alegre, RS, Brazil). Table 9.2 lists what is known about the components of these sealers.

Table 9.2 Composition of Tricalcium Silicate Root Canal Sealers

Product	Manufacturer	System	Composition	Radiopacifier
NeoMTA Plus	Nu Smile, Houston, TX, USA	Powder/water-based gel	Tri/dicalcium silicate	Ta_2O_5
NeoSealer		Organic liquid paste in Syringe	Tri/dicalcium silicate	Ta_2O_5
BioRoot RCS	Septodont, Saint-Maur-des- Fossés, France	Powder/water-based liquid	Tri/dicalcium silicate	ZrO_2
iRoot, EndoSequence BC, Edge Endo, TotalFill BC sealer	Innovative BioCeramix Inc.; Brasseler USA	Organic liquid paste in syringe	Tri/dicalcium silicate	ZrO_2, Ta_2O_5
Endo CPM sealer	EGEO SRL Buenos Aires, Argentina	Powder/liquid	Tri/dicalcium silicate	Bi_2O_3, $BaSO_4$
MTA Fillapex	Ângelus Indústria de Produtos Odontlógicos Ltda	Dual paste	Disalicylate resins	Bi_2O_3
Bio-C		Organic liquid paste in syringe	Tri/dicalcium silicate	ZrO_2
Seal Plus BC	MK Life, Porto Alegre, RS, Brazil	Organic liquid paste in syringe	Tri/dicalcium silicate	ZrO_2

iRoot SP (EndoSequence BC Sealer), TotalFill Sealer, and Bio-C Sealer are premixed sealers. MTA Fillapex and Seal Plus BC are available as two paste syringes that are combined in a mixing tip. BioRoot RCS, Endo CPM Sealer, and MTA Plus sealers are powder/gel systems that the user can mix to the consistency desired. The available materials' manufacturers descriptions are vague. For example, MTA Fillapex has "natural or diluent resins," with "filler and thickening agents." However, what is believed to be in common is that tricalcium silicate powder is a component of all these sealers.

The MTA Fillapex and iRoot SP sealers have nonaqueous vehicles in which the powders are dispersed. For the tricalcium silicate particles to contribute to sealing, they must become hydrated in the tooth by exchange of the nonaqueous vehicle for water in the RCS. EndoCPM Sealer and MTA Plus sealer rely on mixing with water-based gels just before use without a nonaqueous resin. The proportions of the tricalcium silicate powders are not known in any of these materials but are most likely to be highest in MTA Plus because no inert liquid vehicle is included. The inert vehicle is the liquid medium in the pastes of MTA Fillapex and iRoot SP that does not react with the powder. The bioactivity of MTA-type materials has made the use of tricalcium silicate powders in sealers of interest. The possibility of overextension past the apex with a bioactive material leads to the supposition that healing in the periapical area would occur more readily with a tricalcium silicate product. However, two of the sealers have the tricalcium silicate powders dispersed in a nonreactive organic medium, which may detract from the potential benefit of the bioactive powder. To date, no peer-reviewed papers have been published to establish the benefits of tricalcium silicate powders in sealers in human clinical studies. All studies and reports were based on in vitro testing and in vivo animal models. Calcium carbonate is added to Endo CPM Sealer to reduce the pH from 12.5 to 10.0 after setting. By doing so, the manufacturer claimed that surface necrosis in contact with the material is reduced, thereby optimizing the conditions for the enzymatic action of alkaline phosphatase.[107]

The physical properties of some of these sealers have been published. Table 9.3 contains values that are known

Table 9.3 Physical Properties of Tricalcium Silicate Root Canal Sealers

Quality	ISO 6876-2001 ADA 57 Requirement	AH Plus*,†	MTA Fillapex†	EndoSequence BC Sealer	MTA Plus White, Gray‡
Flow (mm)	>20	21, 40	25, 29	23	31
Film thickness (μm)	<50	16	24	22	47
Solubility	<3%	0.1	1.1	2.9	1.9
Dimensional stability	−1% to +0.1%	−0.04	−0.7	+0.09	+0.05
Radiopacity	>3 mm Al	9	4	4	4.9, 5.5
Working time (hours)	NR	4:00, 5:30	0:45, 0:30	2:40	0:20
Setting time (hours)	NR	11:30, 5:10	2:30, 2:15	>24	10:00
Antimicrobial Efficacy	NR	§	§	§	§

*An epoxy resin-based sealer, included for comparative purposes.
†Measured by two authors.
‡If one value is given, then white and gray versions of MTA Plus are equal.
§Antimicrobial effect against the *Enterococcus faecalis*.
MTA, Mineral trioxide aggregate; *NR*, no requirement.

for comparison to the ISO 6876/ADA 57 methods and requirements and the AH Plus sealer. The flow, film thicknesses, solubility, dimensional stability, and radiopacity of the new sealers meet the ISO 6876 requirements. The flow of these sealers varies, but so do the values for AH Plus. The dimensional stability is excellent. The solubility of the sealers is higher than AH Plus, which may be attributed to the formation of partially soluble $Ca(OH)_2$ within the set tricalcium silicate. The radiopacities are less than AH Plus, which is known for its radiopacity.

The working and setting times of the iRoot SP and MTA Plus materials are longer than the MTA Fillapex sealer. iRoot SP has the longest setting time and requires diffusion of water from the dentinal tubules into the sealer to set.[188] In a test where water was added to the EndoSequence BC sealer (iRoot SP), the setting time was reduced to about 150 hours but the hardness was significantly diminished.[168] The EndoSequence BC sealer has been tested for strengthening teeth when used with glass ionomer-coated cones and has been found to be stronger in vitro.[102]

Biocompatibility tests have been performed with some of these sealers. EndoSequence BC sealer was more cytotoxic than AH Plus root canal sealer. The time to become noncytotoxic as judged from mitochondrial enzymatic activity using succinic dehydrogenase was 5 weeks for EndoSequence BC sealer and 3 weeks for AH Plus sealer.[168] Another study reported that, after setting, the cytotoxicity of MTA Fillapex decreased and the sealer presented suitable bioactivity to stimulate hydroxyapatite nucleation.[61,245] A calcium silicate-based sealer possesses a higher modulus of elasticity than AH Plus sealer, unmodified and modified GP cones, and provided better adhesion and sealer penetration at the apical area.[206] A recent systematic review evaluated the biocompatibility of different bioceramic materials in both human and animal mesenchymal cells and reported that all tricalcium silicate materials have similar biological properties to those of MTA, including low cytotoxicity, promotion of cell adhesion and proliferation, and reduced inflammatory cytokines, cells, and mediations.[76] Thus, given the similarity, the choice of bioceramic material should be a professional's decision.

To date, no peer-reviewed papers have been published to establish the benefits of tricalcium silicate powders in sealers in human clinical studies. All studies and reports were based on in vitro testing and in vivo animal models. Despite the lack of well-designed long-term clinical trials, the premixed calcium silicate-based sealers show good physicochemical and biological properties in vitro, with results similar or better than conventional root canal sealer sealers, as observed in in vitro and in vivo animal studies.[268]

A concern associated with the use of tri/dicalcium silicate sealers is the potential difficulty in retreating a hard material.[126,150] It has to be remembered though that, similar to the recommendations for other root canal sealers, a tri/dicalcium sealer is supposed to be employed as a thin film instead of being used en bloc, for both single cone and lateral compaction techniques. For retreatment of root canals that had been filled with GP and tri/dicalcium silicate sealers, removal of the sealer remnants and the regaining of apical patency were no worse than with epoxy or resin-based sealers.[57,192]

MEDICATED SEALERS

Sealers containing paraformaldehyde are strongly contraindicated in endodontic treatment (Fig. 9.20). Although the lead and mercury components may have been removed from these zinc oxide–eugenol formulations over time, the severely toxic paraformaldehyde content has remained a constant. These sealers are not approved by the U.S. Food and Drug Administration[1] and are unacceptable under any circumstances in clinical treatment because of the severe and permanent toxic effects on periradicular tissues.[263]

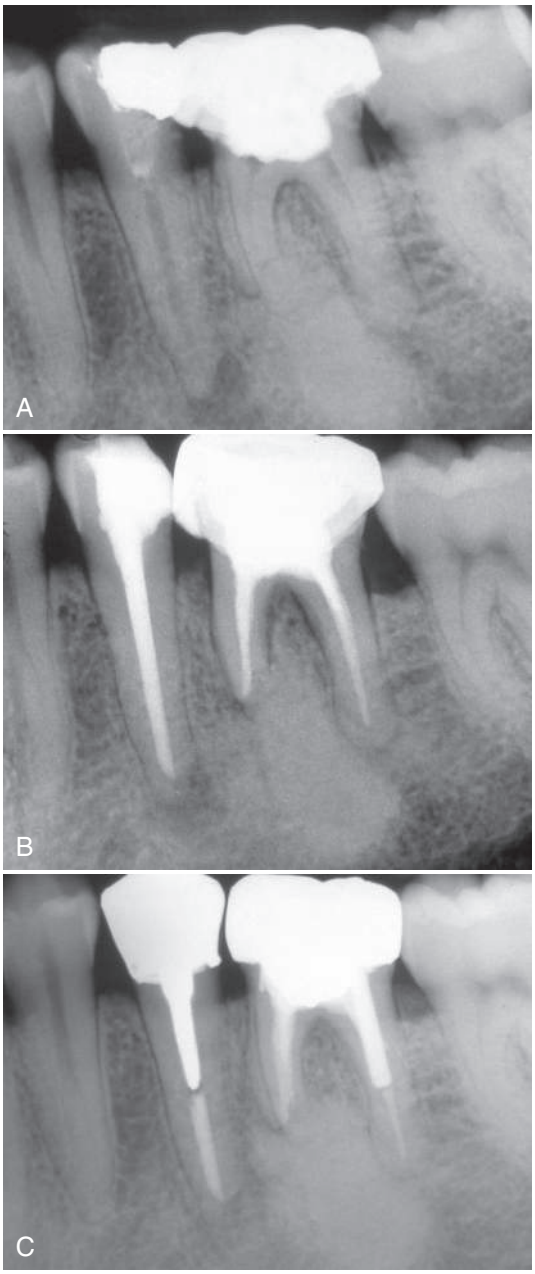


Fig. 9.20 Patient treated with Sargenti paste in her mandibular left second premolar and first molar. **A,** Pretreatment radiograph exhibits an osteolytic response associated with the premolar and a proliferative response associated with the molar. **B,** Posttreatment radiograph of the teeth. **C,** One-year follow-up radiograph exhibiting osseous regeneration apical to the second premolar.

A paste containing 6.5% paraformaldehyde as well as lead and mercury was advocated for use by Sargenti[248] and originally marketed as N-2. Lead has been reported in distant organ systems when N2 is placed within the radicular space.[207] In another study the investigators reported the same results regarding systemic distribution of the paraformaldehyde component of N2.[32] Removal of the heavy metals resulted in a new formulation: RC2B. Other paraformaldehyde sealers include Endomethasone, SPAD, and Reibler's paste. The toxic in vivo effects of these materials on the pulp and periapical tissues have been demonstrated over time.[64,193]

In addition to the toxic nature of the material, clinicians employing the material place it with a lentulo spiral. Overextension has resulted in osteomyelitis and paresthesia,[88] where irreversible neurotoxicity has manifested as dysesthesia as a result of paraformaldehyde being forced through the apical foramen into the periapical tissues.[148] The reader is referred to Chapter 27 for further discussion about this harmful material and technique.

Sealer Placement

Various methods of sealer placement have been advocated, including a master cone, lentulo spiral, a file and/or reamer, and ultrasonics. Investigators compared sealer placement using a file rotated counterclockwise, the lentulo spiral (Fig. 9.21), an ultrasonic file, and coating the master GP cone.[306] Placement did not differ with the various techniques. However, the investigators noted the most variation in sealer coating was in the apical area.[306] Another study compared sealer placement with a K-type file, the lentulo spiral, and using the master cone in curved canals. Results demonstrated no significant differences in the techniques after obturation. No technique covered more than 62.5% of the RCS.[118] Other investigators reported that ultrasonics produced the best sealer distribution when used circumferentially.[3,279] In the past, it has been presumed that instrumented RCS need to be thoroughly dried prior to obturation with root filling materials and sealers. Indeed, there were leakage studies reporting that a better apical seal might be achieved by drying the canals with 95% ethanol prior to obturation.[280] A recent study reported that, for some sealers (iRoot SP, AH Plus, Epiphany, and MTA Fillapex), better adhesion to the dentinal walls might be achieved by leaving the canals slightly moist. This can be accomplished by drying the RCS with low vacuum via a Luer adapter for 5 seconds followed by insertion of a paper point for 1 second.[188] Similar results were also reported with the use of the EndoREZ sealer.[324] This, however, does not mean that a RCS should remain totally flooded with water prior to obturation because such a level of residual moisture would adversely affect the adhesive of those root canal sealers to intraradicular dentin.[188]

The method of obturation does not seem to affect the sealer distribution on the canal wall in the apical portion of the canal. However, lateral compaction results in better distribution in the midcoronal areas when compared with warm vertical compaction.[312] Another well-controlled study reported that none of five evaluated obturation techniques resulted in uniform sealer distribution along the entire length of the core obturation material.[130] Evidence indicates that the method of obturation affects the sealer penetration into tubules. This was exemplified by a study reporting that thermoplastic techniques produced deeper sealer penetration into tubules.[74] Removal of the smear layer enhances sealer penetration into the dentinal tubules.[152]

Core Materials

Although a variety of core materials have been used in conjunction with a sealer/cement, the most common method of obturation involves GP as a core material. Regardless of the obturating technique, emphasis should be placed on the process of C&S the RCS. The materials and techniques described do not routinely create an impervious seal of the RCS. All materials leak to some extent.[4] The choice of the obturation technique(s) depends on the unique circumstances with which each case presents.

The properties of an ideal obturation material have been outlined by Grossman (Box 9.2).[110] Historically, a variety of materials have been employed to obturate the RCS. Solids, semisolid materials, and pastes have been employed. A common solid material used in the past was the silver cone.

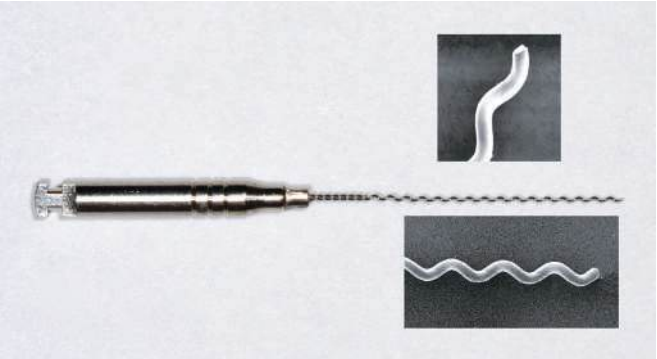


Fig. 9.21 Lentulo spiral used for sealer placement during obturation.

Box 9.2 Properties of an Ideal Obturation Material

- Easily manipulated and provides ample working time
- Dimensionally stable with no shrinkage once inserted
- Seals the canal laterally and apically, conforming to its complex internal anatomy
- Nonirritating to the periapical tissues
- Impervious to moisture and nonporous
- Unaffected by tissue fluids—no corrosion or oxidization
- Inhibits bacterial growth
- Radiopaque and easily discernible on radiographs
- Does not discolor tooth structure
- Sterile
- Easily removed from the canal if necessary

SILVER CONES

Jasper introduced cones made of silver, which he claimed produced the same success rate as GP and were easier to use.[136] The rigidity provided by the silver cones made them easier to place and permitted more predictable length control; however, their inability to fill the irregularly shaped RCS permitted leakage (Fig. 9.22). Silver cones were believed to possess an oligodynamic property which, if true, would have resulted in the destruction of microbes within the RCS. Unfortunately, it was not true. Moreover, when silver points contact tissue fluids or saliva, they corrode.[38] The corrosion of such products has been found to be cytotoxic and produces pathosis or impedes periapical healing.[260]

With the introduction of rigid silver cones, it became possible to easily place them to length. This resulted in clinicians often failing to properly clean and shape RCS before obturation. Treatment failures were the result of leakage and failure to remove the irritants from the RCS. The use of silver cones today is considered to be below the standard of care in contemporary endodontic practice. For further information about this material and technique, the reader is referred to Chapter 27.

GUTTA-PERCHA

GP is the most popular core material used for obturation. Major advantages of GP are its plasticity, ease of manipulation, minimal toxicity, radiopacity, and ease of removal with heat or solvents. Disadvantages include its lack of adhesion to dentin and, when heated, shrinkage upon cooling. GP is the *trans*-isomer of polyisoprene, natural rubber, and exists in two crystalline forms, α and β.[108] In the unheated β phase, the material is a solid mass that is compactable. When heated, the material changes to the α phase and becomes pliable and tacky and can be made to flow when pressure is applied. A disadvantage to the α phase is that the material shrinks on setting.[257]

GP cones consist of approximately 20% GP, 65% zinc oxide, 10% radiopacifiers, and 5% plasticizers.[96] Attempts have been made to make GP more antimicrobial, but the clinical effectiveness of adding these materials has not been demonstrated. Moreover, to exert an antimicrobial pharmacological effect, the active ingredient must leach out of the GP, which could have a detrimental effect on long-term sealability.

Unlike rubber, room temperature GP cannot be compacted or made to flow. Compaction results in transmission of forces to the material and the canal wall equally and may

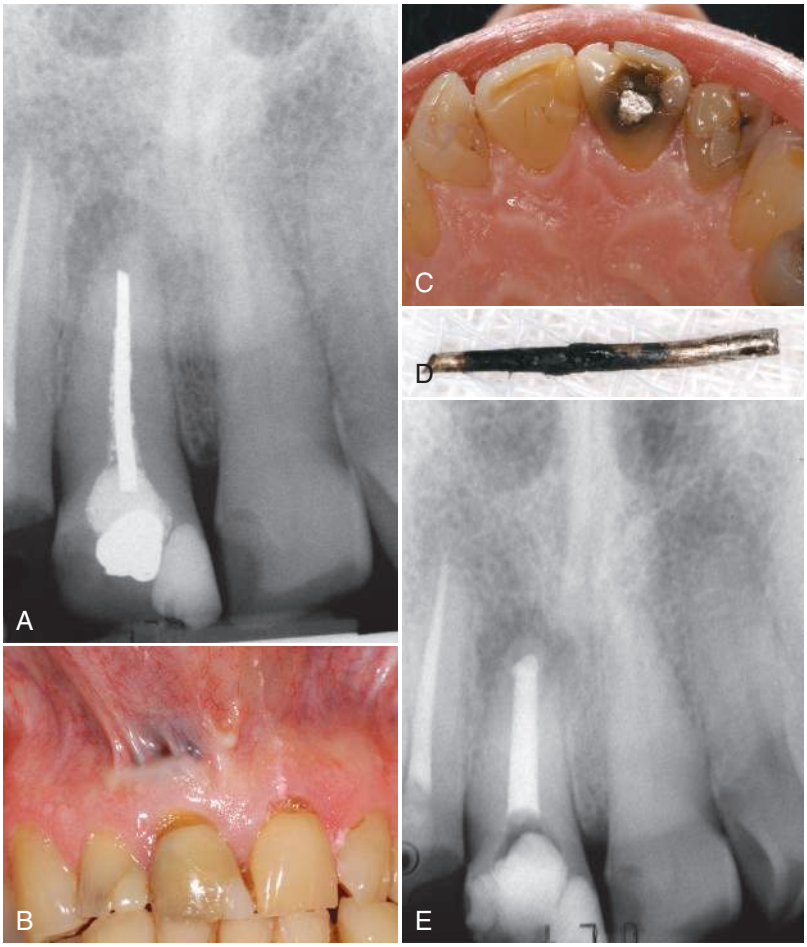


Fig. 9.22 Silver cones have been advocated for ease of placement and length control. **A,** Radiograph of a maxillary right central incisor obturated with a silver cone. **B,** Tissue discoloration indicating corrosion and leakage. **C,** Lingual view indicates coronal leakage. **D,** Corroded silver cone removed from the tooth. **E,** Posttreatment radiograph of the tooth.

result in root fracture. GP can be made to flow if it is modified by either heat or solvents such as chloroform. This permits adaptation to the irregularities of the canal walls.

The α form of GP melts when heated above 65°C. When cooled extremely slowly, the α form will recrystallize. Routine cooling results in the recrystallization of the β form. Although the mechanical properties for the two forms are the same, when α-phase GP is heated and cooled, it undergoes less shrinkage, making it more dimensionally stable for thermoplasticized techniques. The use of α-phase GP for obturation has increased as thermoplastic techniques have become more common.

GP cones are available in standardized and nonstandardized sizes. Standardized sizes conform to requirements contained in specification published by the International Organization of Standardization (ISO) or ADA American National Standards Institute (ADA ANSI). The nonstandard nomenclature refers to the dimensions of the tip and body (Fig. 9.23). A fine-medium cone has a fine tip with a medium body. Standardized cones are designed to match the taper of stainless steel and nickel titanium instruments (Figs. 9.24 and 9.25). A size 40, 0.04 cone has a tip of 0.4 mm and a taper of 0.04 mm per millimeter. Unfortunately, ISO and ADA ANSI standards allow tolerances and, coupled with less-than-completely accurate manufacturing processes, the actual cone size varies as does the tip and taper of the master apical file.[106]

Although GP points cannot be heat sterilized, a study reported that GP points can be sterilized by placing in 5.25% NaOCl for 1 minute. This study also reported that 2% glutaraldehyde, 2% chlorhexidine, and 70% ethyl alcohol were not effective in killing *Bacillus subtilis* spores, one of the hallmark organisms used for testing effective antimicrobial effectiveness.[270]

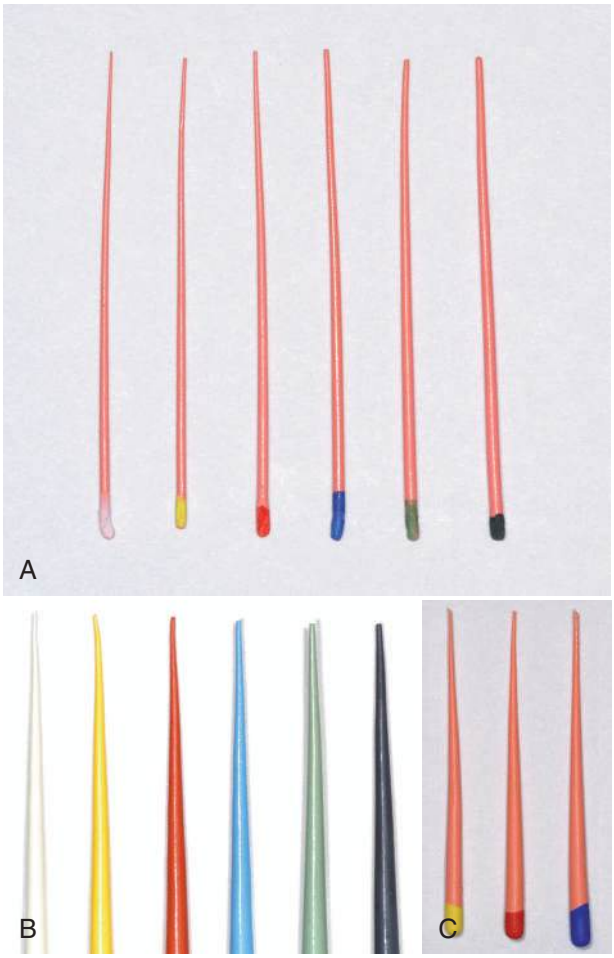


Fig. 9.24 **A,** Standard GP cone sizes #15 to #40. **B,** Standard cones #.06, taper sizes #15 to #40. **C,** Standard cones ProTaper F1, F2, and F3.

ACTIV GP

Activ GP (Brasseler USA) consists of GP cones impregnated on the external surface with glass ionomer (see Fig. 7.14). Single cones are used with a glass ionomer sealer. Available in 0.04 and 0.06 taper cones, the sizes are laser-verified to

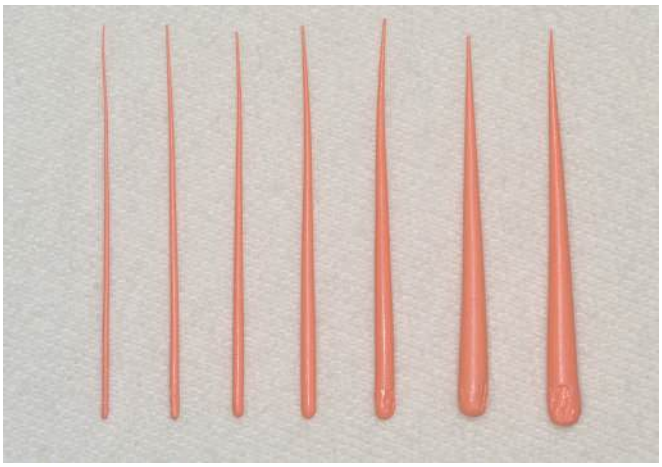


Fig. 9.23 Nonstandard GP cones: extra fine, fine fine, fine, medium fine, fine medium, medium, large, and extra-large.

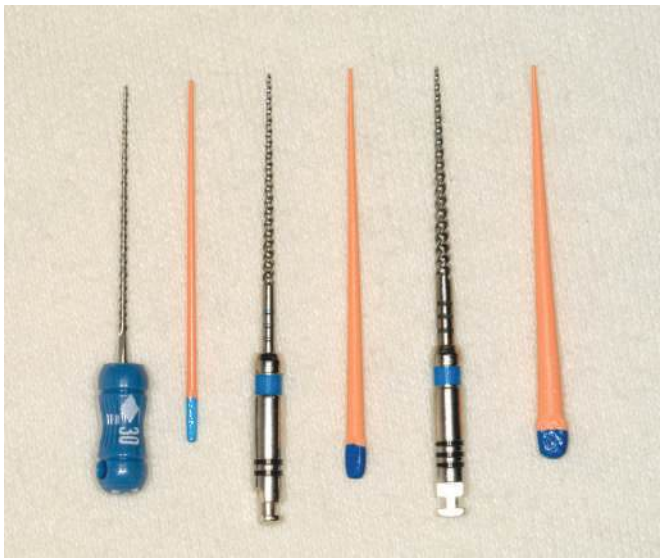


Fig. 9.25 Size #30 standard GP points exhibiting #.02, #.04, and #.06 tapers.

help ensure a more precise fit. The single cone technique is designed to provide a bond between the dentinal canal wall and the master cone. A bacterial leakage study comparing Activ GP/glass ionomer sealer, Resilon/Epiphany, and GP/AH Plus demonstrated no statistically significant differences in leakage at 65 days.[95]

CUSTOM CONES

When the apical foramen is excessively large or the prepared RCS is large, a custom cone may need to be fabricated (Fig. 9.26). This permits the adaptation of the cone to the canal walls, reduces the potential for extrusion of the core material, and may improve the resultant seal.[26,180] The technique involves selection of a master cone and fitting that cone 2 to 4 mm short of the prepared length with frictional resistance. The cone is grasped with locking cotton pliers or a hemostat so that it can be placed into the RCS in the same spatial relationship each time. The tip of the GP cone is removed and the resultant new tip is softened in chloroform, eucalyptol, or halothane for 1 or 2 seconds, depending on the clinical requirements. Only the outer superficial portion of the cone is softened. The central core of the cone should remain semirigid. The cone is then placed back into the RCS to the working length. The process can be repeated until an adequate impression of the canal is obtained at the prepared length. A radiograph is exposed to verify proper fit and position. An alternative to solvents is softening with heat.[145]

A large RCS may necessitate the custom fabrication of a large master cone before canal adaptation. This may be accomplished by heating two, or more, large GP cones and rolling the mass between two glass slabs until an appropriate size is obtained (Fig. 9.27). A spatula may also be used to shape the cone.

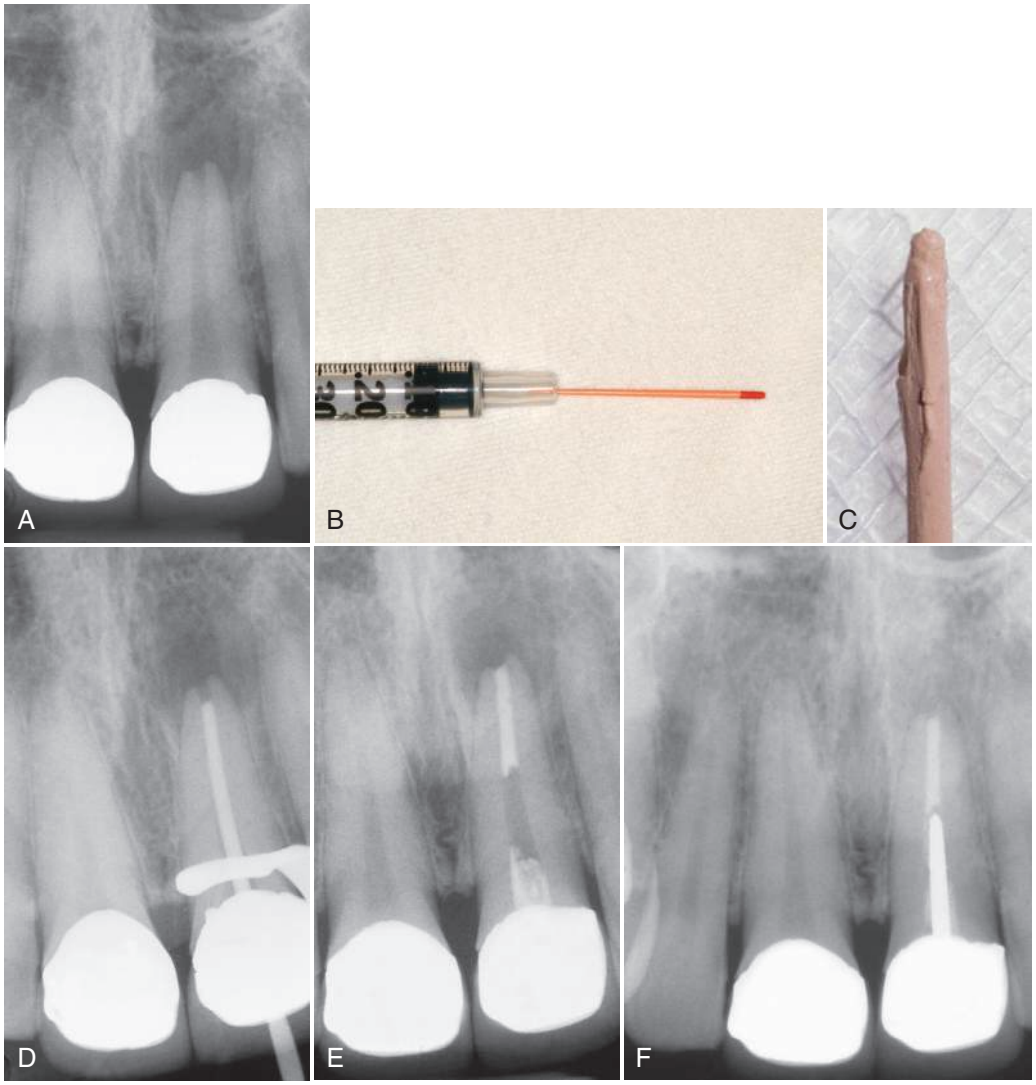


Fig. 9.26 Apical root resorption often results in an open apex requiring fabrication of a custom cone. **A,** Pretreatment radiograph of a maxillary left central incisor with pulp necrosis and chronic apical periodontitis. Apical root resorption is present. **B,** In fabricating a custom master cone, a GP point is fit several millimeters short before softening in solvent and tamping into place. **C,** Softening the apical 2 to 3 mm in chloroform that has been placed in a tuberculin syringe. **D,** The completed custom cone represents an impression of the apical portion of the RCS. **E,** The posttreatment radiograph with post space prepared. **F,** A 1-year follow-up radiograph demonstrating osseous regeneration.

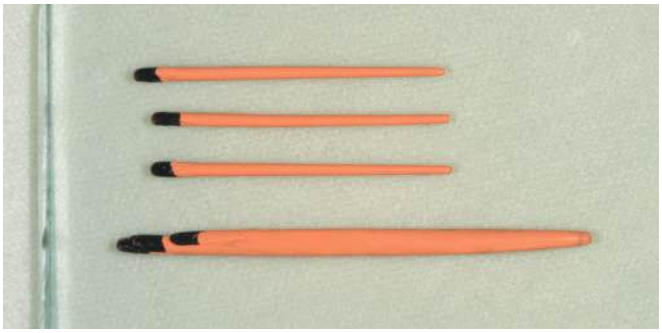


Fig. 9.27 For large RCS, several GP points can be heated and rolled together, using a spatula or two glass slabs.

Methods of Obturation

To date, little evidence exists to support one method of obturation as being superior to another based on outcome assessment investigations.[14,195] The prospective Toronto studies have suggested that warm vertical compaction may be superior to lateral compaction.[73] However, definitive evidence is lacking.[218]

LATERAL COMPACTION

Lateral compaction is a common method for obturation (Figs. 9.28 and 9.29).[50] The technique can be used in most

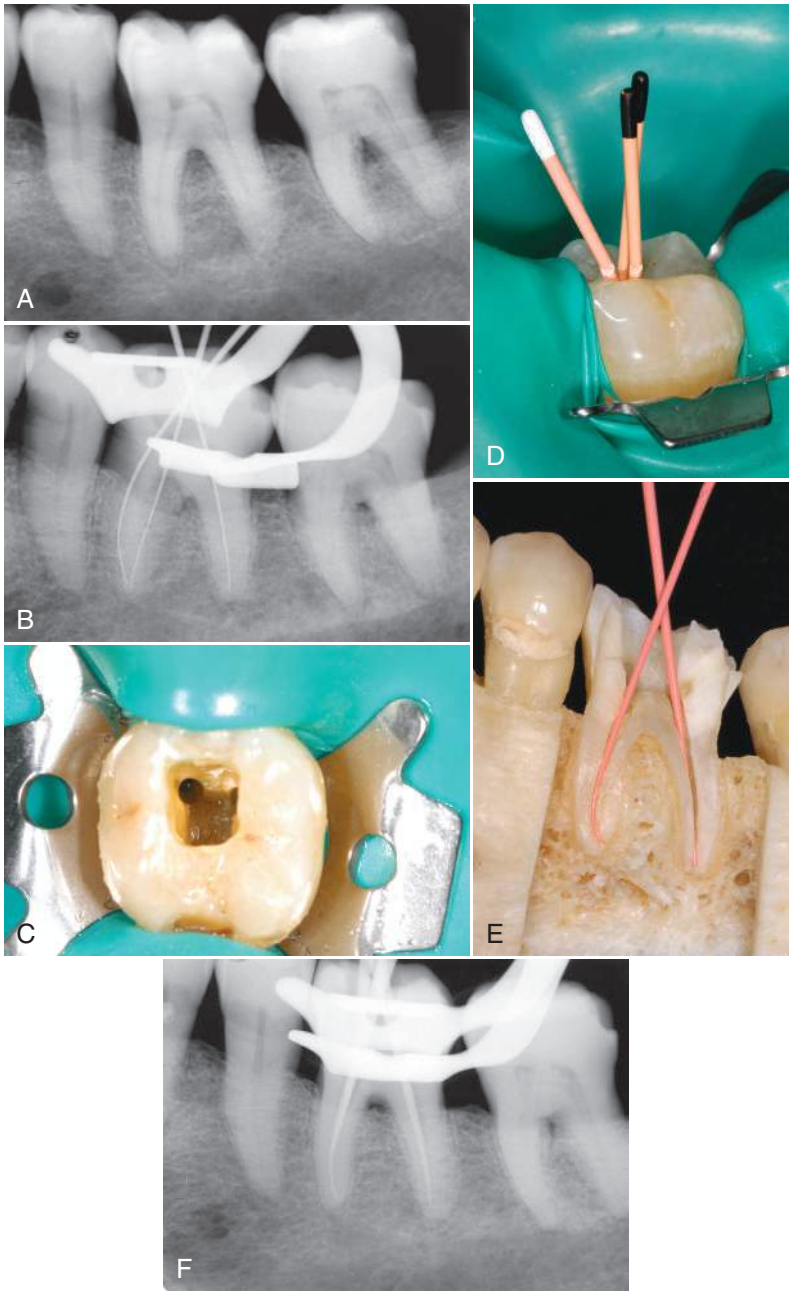


Fig. 9.28 Left first mandibular molar. **A,** Pretreatment radiograph. **B,** Working length radiograph. **C,** Coronal access opening, demonstrating the prepared mesiobuccal canal. **D,** Standardized master cones with coronal reference marked. **E,** Standard master cones fit to length as they exhibit minimal taper and permit deeper penetration of the spreader. **F,** Master cone radiograph.

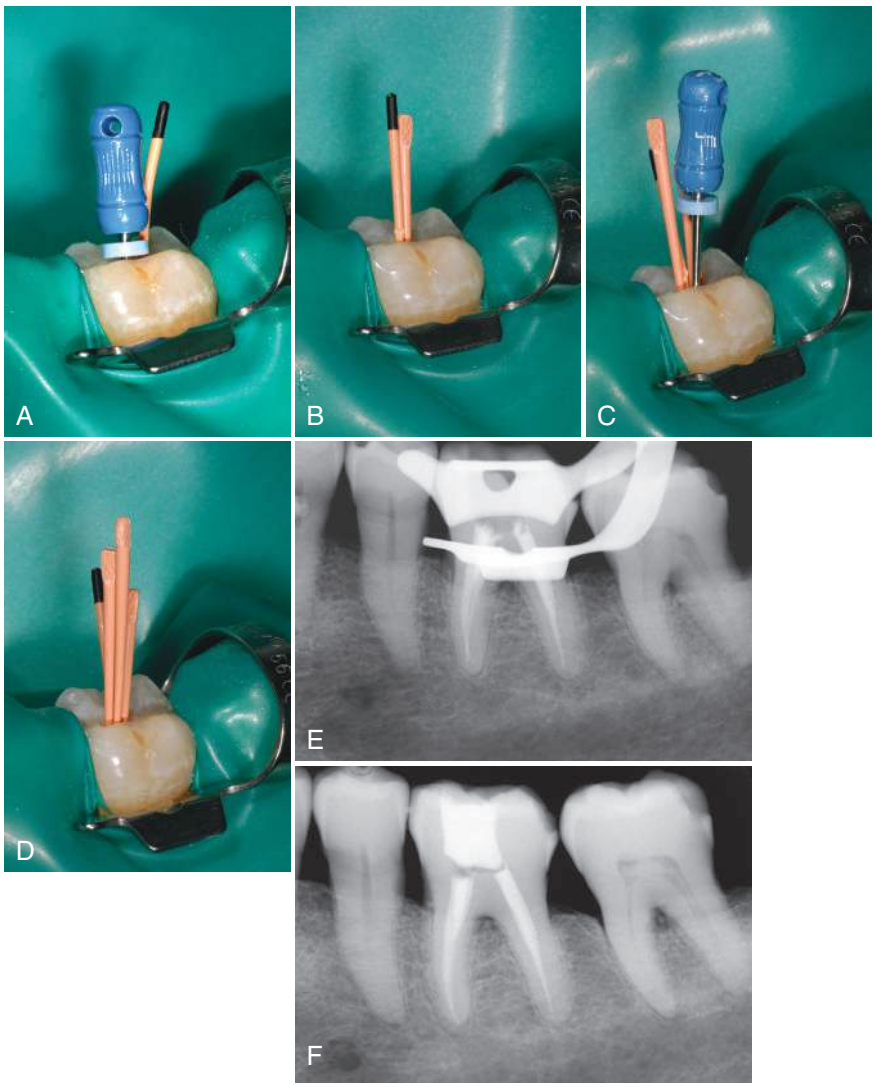


Fig. 9.29 Lateral compaction. **A,** Finger spreader in place. **B,** Fine-medium accessory cone placed in the space created by the spreader. **C,** Finger spreader placed in preparation, creating space for additional accessory cones. **D,** Additional cones are placed until the spreader does not penetrate past the coronal one third of the canal. The cones are then removed at the orifice with heat, and the coronal mass is vertically compacted with a plugger. **E,** Interim radiograph may be exposed to assess the quality of obturation. **F,** Posttreatment radiograph demonstrating adequate length, density, and taper. The gutta-percha is removed to the level of the orifice, and a coronal seal has been established with an adequate provisional restoration.

clinical situations and provides for predictable length control during compaction.[103] A disadvantage is that the technique may not fill canal irregularities[315] as well as warm vertical compaction or other thermoplastic techniques.[309] The procedure can be accomplished with any of the acceptable sealers.

After RCS preparation, a standard cone is selected that has a tip and taper consistent with the prepared canal diameter at the working length established by the master apical file. Standard cones generally have less taper when compared with nonstandard cones and will permit deeper spreader penetration, which will result in a better-quality resultant seal.[220] An alternative is to adapt an appropriately tapered nonstandard cone by cutting small increments from the tip. This "master cone" is measured and grasped with forceps so that the distance from the tip of the cone to the reference point on the forceps is equal to the prepared length. A reference point on the cone can be made by pinching the cone. The cone is placed into the RCS and, if an appropriate size is selected, there will be resistance to displacement or "tug back." If the cone is loose, it can be adapted by removing small increments from the tip. If the master cone fails to go to the prepared length a smaller cone can be selected. Devices are available to cut cones accurately at a predetermined length (Tip Snip; SybronEndo). When the cone extends beyond the prepared length, a larger cone must be adapted or the existing cone shortened until there is resistance to displacement at the corrected working length.

The master cone placement is confirmed with a radiograph. The canal is irrigated and dried with paper points. Sealer is applied to the canal walls and a spreader is prefit so as to allow it to be inserted to within 1 to 2 mm of the working length. Appropriately sized accessory points are also selected to closely match the size of the spreader to be used. The correlation between spreader size and nonstandard cones is variable,[43,322]

Finger spreaders may provide better tactile sensation and are less likely to induce fractures in the root when compared with the more traditional D-11T hand spreader.[164] In addition to the type of spreader, forces applied, and amount of dentin removed, spreader size may be a factor in root fracture, with large sizes inducing more stress.[225] Spreaders made from nickel titanium are available and provide increased flexibility,[30] reduced stress,[101] and deeper penetration when compared with stainless steel instruments.[258] The spreader should fit to within 1 to 2 mm of the prepared length, and, when introduced into the RCS with the master cone in place, it should be within 2 mm of the working length.[8] There appears to be a correlation between the establishment of a higher quality seal and the depth of spreader penetration.[8]

After the spreader has been placed to its maximum depth, it is removed by rotating it back-and-forth as it is withdrawn. An accessory cone is placed in the space vacated by the spreader. The process is repeated until the spreader no longer goes beyond the coronal one third of the canal. The excess GP is removed with heat and the coronal mass is compacted with an appropriate unheated plugger. Only light pressure is required during lateral compaction because the GP is not condensed, and because as little as 1.5 kg of pressure is capable of fracturing a root (Fig. 9.30). In addition to the force applied, investigators have noted that removal of excessive amounts of dentin during preparation is a significant factor in root fracture.[307]

A disadvantage to lateral compaction is that the process does not produce a homogeneous mass. The accessory and master cones are laminated and remain separate. It is envisaged that the space between each of the cones is filled with sealer to aid in establishment of a water-tight seal

The excess GP in the chamber is then seared off and vertically compacted with a heated plugger at the orifice or approximately 1 mm below the orifice in posterior teeth. In anterior teeth, the desired level is the cementoenamel junction on the facial surface to avoid aesthetic issues if the dentin should become stained. An alternative to lateral compaction with finger spreaders is the use of ultrasonics and a clinical study reported a 93% success rate.[321]

WARM VERTICAL COMPACTION

Schilder introduced warm vertical compaction of GP as a method of obturating the radicular space in three dimensions.[256] Preparation requirements for the technique include preparing an RCS with a continuously tapering funnel and keeping the apical foramen as small as possible.

Various ISO standardized instruments are also available (Fig. 9.31).

The technique involves fitting a master cone 0.5 to 2 mm short of the prepared working length with resistance to displacement (Fig. 9.32). This ensures that the cone diameter is larger than the prepared canal at the terminus. Nonstandard cones that closely replicate the canal taper are best because they permit the development of hydraulic pressure during compaction. After the adaptation of the master cone, it is removed and sealer is applied to the cone and the walls of the prepared canal. The cone is placed in the canal and the coronal portion is removed with a heated instrument. A heated spreader or plugger is used to remove portions of the coronal GP in successive increments and soften the remaining material in the canal. The Touch 'n Heat (Kerr) (Fig. 9.33) and System B (Kerr Dental) (Fig. 9.34) are alternatives to applying heat with a flame-heated instrument because they permit improved temperature control. A plugger is inserted into the RCS and the GP is compacted, forcing the plasticized material apically. The process is repeated until the apical portion has been reached. The coronal RCS is back-filled using small preheated pieces of GP. The sectional method consists of placing 3 to 4 mm sections of GP approximating the size of the canal into the root, applying heat, and compacting the mass with a plugger.

One study measured temperature changes in an RCS with warm vertical compaction.[34] The maximal temperatures occurred coronally and decreased apically. The authors reported that the maximal temperature in the canal was 118°C 8 mm from the apex. At 0 to 2 mm from the apical terminus, the maximal temperature had decreased to 44°C. Another study compared root surface temperatures

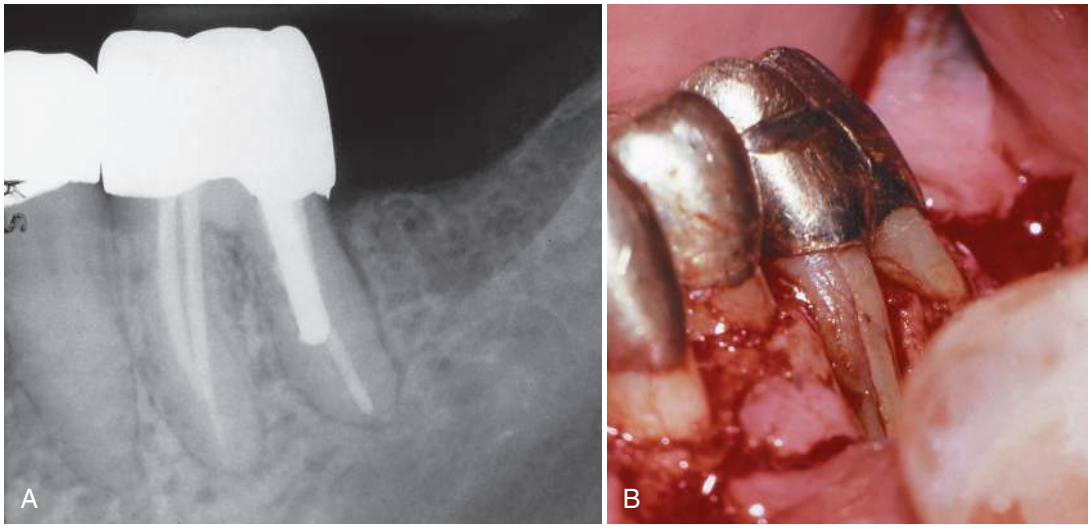


Fig. 9.30 Vertical root fractures can occur with excessive compaction forces. **A,** Follow-up radiograph of a mandibular left first molar. A deep isolated periodontal probing defect was associated with the buccal aspect of the mesiobuccal root. **B,** Flap reflection revealed a vertical root fracture.

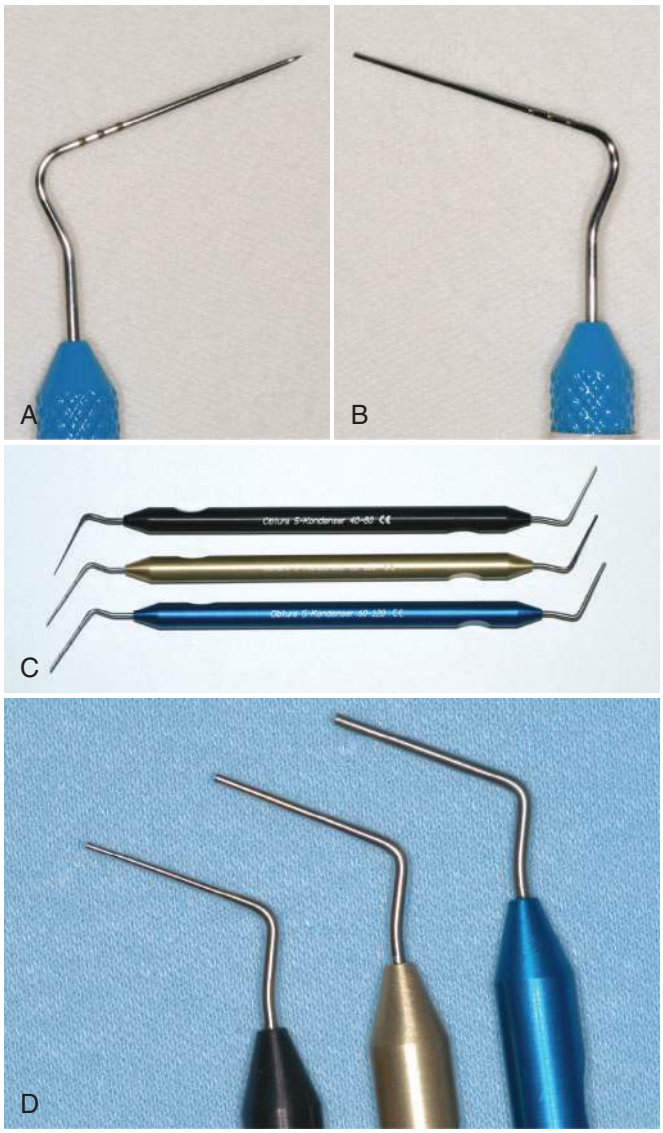


Fig. 9.31 Various pluggers are manufactured for compacting warm GP. **A,** ISO standard spreader. **B,** ISO standard plugger. **C,** Obtura S Kondensers. **D,** Obtura S-Kondensers. (***C***, Courtesy Obtura Spartan, Earth City, MO.)

for warm vertical obturation using the System B heat source, the Touch 'n Heat device, and a flame-heated carrier in maxillary and mandibular incisors and premolars 2 mm below the cementoenamel junction. System B and the Touch 'n Heat produced a surface temperature rise that was less than 10°C for all maxillary incisors and premolar teeth. The Touch 'n Heat produced a greater than 10°C rise in mandibular incisors. The flame-heated carrier produced temperature changes greater than 10°C in all experimental teeth. Because the critical level of root surface heat required to produce irreversible bone damage is believed to be greater than 10°C, the findings suggest that warm vertical compaction with the System B should not damage supporting periodontal structures. However, caution should be exercised with the Touch 'n Heat and flame-heated carriers.[161] The flame-heated spreader has no heat controls and the Touch 'n Heat controls only send heat to the tip and receive no feedback from the tip based on clinical conditions. The System B has a built-in computer that allows it to receive feedback from the tip and adjust the tip temperature based on the clinical conditions, which allow the tip to remain at the dialed-in temperature.

The potential for vertical root fracture is also present with warm vertical compaction.[33] The forces developed appear to be equal to lateral compaction. Investigators compared warm vertical compaction and lateral compaction as a function of time. Results indicated that the forces developed with the two techniques were not significantly different. In a follow-up study, the mean value for wedging with warm vertical compaction was 0.65 ± 0.07 kg, whereas for lateral compaction it was 0.8 ± 0.1 kg.

Warm thermoplastic techniques have the advantage of producing movement of the plasticized GP within the obturated RCS, creating a more homogeneous mass of resultant GP and filling irregularities and accessory canals better than lateral compaction.[311] This was illustrated in a study that reported a correlation between the quality of adaptation and the depth of heat application and canal size. Heat application close to the apical extent of the preparation produced the best results, and adaptation was better in small RCS when compared with wider ones.[310] However, thermoplasticized techniques resulted in more extrusion of obturating materials.[154] There appeared to be no consistent differences between the techniques in sealing the RCS.[311] Warm vertical compaction also influenced setting time and flow of sealers.[228]

Advantages of warm vertical compaction include filling of canal irregularities and accessory canals. Disadvantages include a slight risk of vertical root fracture because of compaction forces, less length control than with lateral compaction, and the potential for overextension of obturation materials into the periradicular tissues. Warm vertical compaction is also difficult in more curved RCS where the rigid pluggers may be unable to penetrate to the necessary depth. Occasionally, the RCS have to be enlarged and tapered more to enable rigid carriers to penetrate within 4 to 5 mm of the working length. Additional removal of RCS dentin can weaken the root, making it more susceptible to root fracture.

CONTINUOUS WAVE COMPACTION TECHNIQUE

A variation of warm vertical compaction is the continuous wave compaction technique.[45] The increasing use of nickel titanium rotary preparation techniques and the fabrication of greater taper standard cones have resulted in more clinicians using thermoplasticizing techniques. The manufacturing of GP cones to mimic the tapered preparation permits the application of greater hydraulic force during compaction when appropriately tapered pluggers are used. The continuous wave compaction technique employs the System B connected to 0.04, 0.06, 0.08, 0.10, or 0.12 tapered stainless steel dead-soft pluggers (see Fig. 9.34). The 0.06 tapered plugger also approximates the fine nonstandard GP cone, while the 0.08 plugger approximates the fine-medium cone, the 0.10 plugger the medium cone, and the 0.12 plugger the medium-large cone. Pluggers are consistent with the ProFile GT instruments (DENTSPLY Sirona), and Autofit GP cones (Kerr Endo) are also available. In summary, after selecting an appropriate master

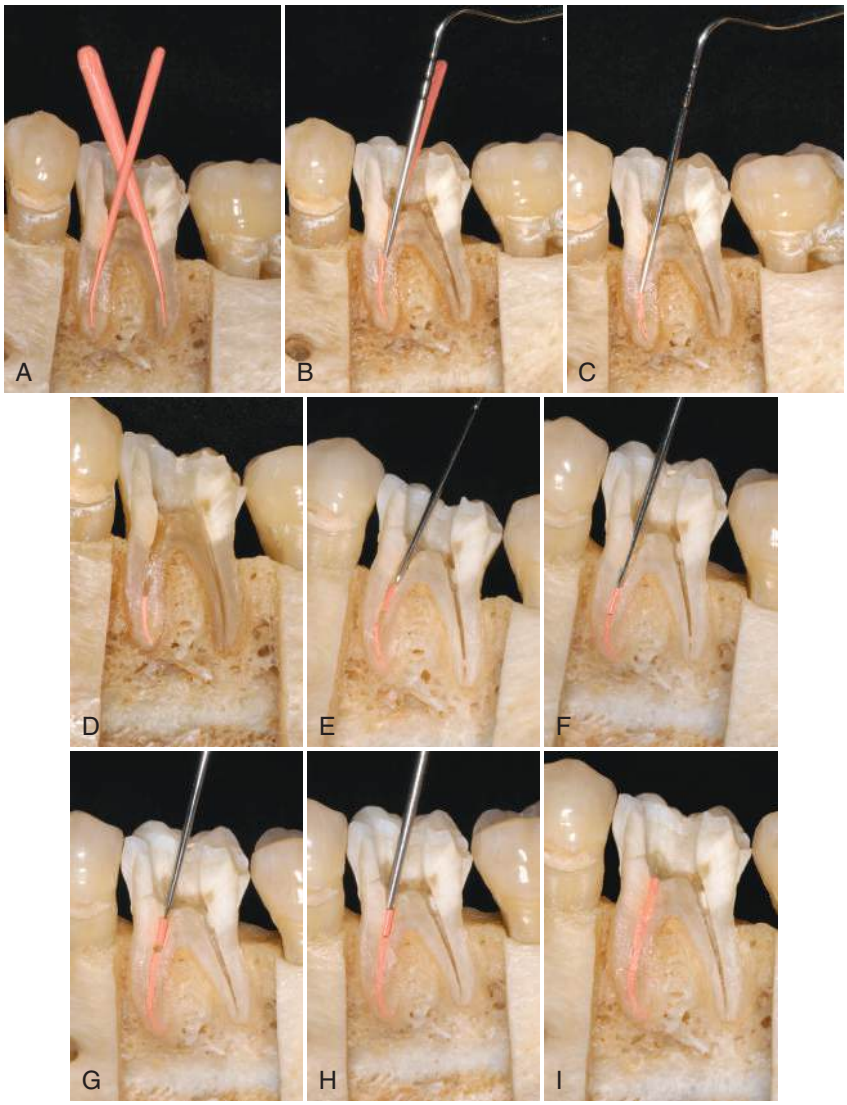


Fig. 9.32 Warm vertical compaction of GP employs heat and various condensers. **A,** Nonstandard cones are selected and fit short of the prepared length because they more closely replicate the prepared canal. **B,** Heated pluggers or spreaders are used to apply heat to the master cone and remove the excess coronal material. **C,** A room temperature plugger is used to compact the heated GP. **D,** Apical compaction is complete. **E,** A GP segment is placed in the canal, and heat is applied. **F,** The heated segment is compacted. **G,** The process is repeated for the coronal portion of the canal by placing and heating a segment of GP. **H,** A plugger is again used to compact the heated material. **I,** Completed obturation.



Fig. 9.33 Touch 'n Heat unit. (Courtesy SybronEndo, Orange, CA.)

cone, a series of pluggers from large to small are prefitted to finally reach within 5 to 7 mm of the prepared length (Fig. 9.35).[45]

The continuous wave and warm backfill technique cause significantly higher temperatures than other obturation techniques by virtue of the technique.[80] This occurs because the plugger of the System B unit is inserted into the canal orifice and activated to remove excess coronal material, while the series of compaction and removal of GP takes place until the plugger is moved rapidly (1 to 2 seconds) to within 3 mm of the binding point (Fig. 9.36). The continuous wave technique was reported to have higher temperatures at 3 mm and 6 mm from the apex than the warm backfill technique.[80]

It should be noted that with the continuous wave technique, the tip of the heated plugger is placed only to within 5 to 7 mm from the tip of the GP. The apical portion of the GP remains essentially a single-cone technique as

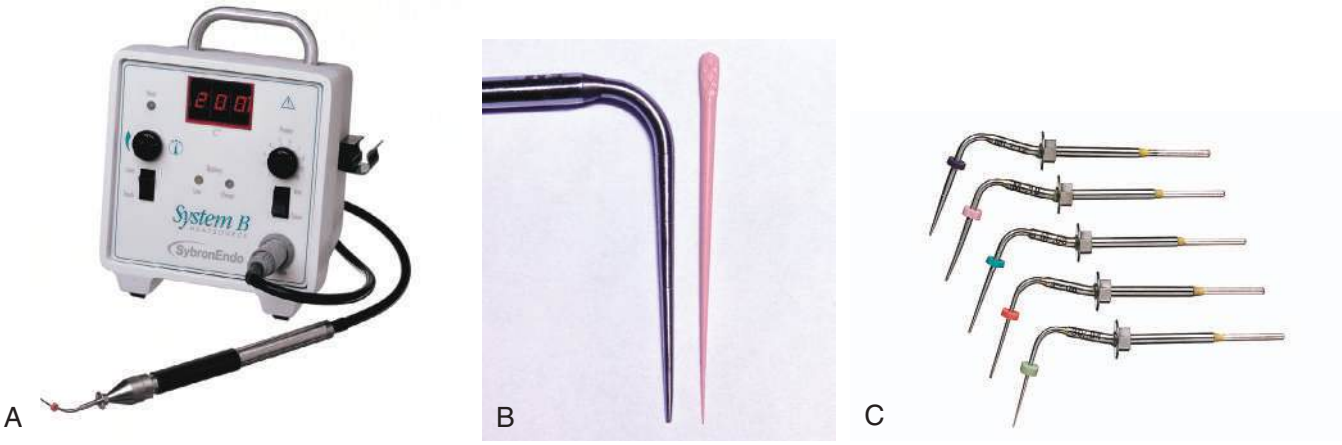


Fig. 9.34 Continuous wave obturation uses the System B unit. **A,** The System B unit. **B,** System B plugger with a nonstandard cone of similar taper. **C,** System B pluggers. (Courtesy SybronEndo, Orange, CA.)



Fig. 9.35 System B plugger fit.

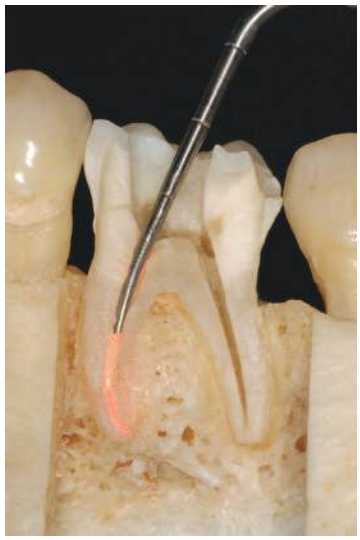


Fig. 9.36 System B activation and compaction.

the heat transfer does not take place in the apical 2 to 5 mm of the GP.[109]

In ovoid canals, where the canal configuration may prevent the generation of hydraulic forces, an accessory cone can be placed alongside the master cone before compaction. With type II canals, the master cones are placed in both canals before compaction. A hand plugger is used to stabilize the cone in one canal while the other is being obturated.

Filling the space left by the plugger may be accomplished by a thermoplastic injection technique (Obtura II or Ultrafil 3D [Coltène/Whaledent], Calamus [DENTSPLY Sirona Dental Specialties], Elements free [Kerr Dental], or HotShot [Discus Dental, Culver City, CA]) (Fig. 9.37)[138] or by fitting an accessory cone into the space with sealer, heating it, and compacting by short applications of heat and vertical pressure.

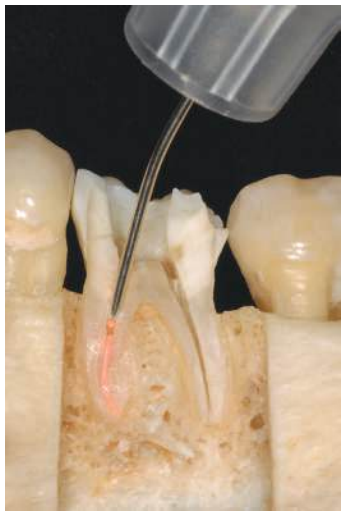


Fig. 9.37 Backfill by the Obtura II thermoplastic injection technique.

THERMOPLASTIC INJECTION TECHNIQUES

Heating of GP outside the tooth and injecting the material into the canal is an additional variation of a thermoplastic technique (Fig. 9.38). The Obtura III (Fig. 9.39), Calamus (Fig. 9.40), Elements (Fig. 9.41), HotShot (Fig. 9.42), and Ultrafil 3D (Fig. 9.43) are available devices. The Obtura II system heats the GP to 160°C, whereas the Ultrafil 3D system employs a low-temperature GP that is heated to 90°C.

Obtura III

The Obtura III system (Obtura Spartan, Earth City, MO, USA) consists of a handheld "gun" that contains a chamber surrounded by a heating element into which pellets of GP are loaded (see Fig. 9.40). Silver needles (varying gauges of 20, 23, and 25) deliver the thermoplasticized material into the RCS. The control unit allows the operator to adjust the temperature and thus the viscosity of the GP. At 6 mm from the apex, one study reported that the highest internal temperature with the Obtura system was 27°C.[283]

Canal preparation is similar for other obturation techniques. The apical terminus should be as small as possible to prevent overextension of GP. The technique requires the use of sealer and, once the canal is dried, the canal walls are coated with sealer using the last file used to length or a paper point. GP is preheated in the gun and the needle is positioned in the RCS so that it reaches within 3 to 5 mm of the apical preparation. GP is then gradually, passively injected by squeezing the trigger of the "gun." The needle backs out of the canal as the apical portion is filled. Room-temperature prefittted pluggers, dipped in alcohol, are used to compact the GP. A segmental technique may also be used, in which 3- to 4-mm segments of GP are sequentially injected and compacted. In either case, compaction should continue until the GP cools and solidifies to compensate for the contraction of the GP that takes place on cooling.

The difficulties with this system include lack of length control. Both overextension and underextension are common results. To overcome this drawback, a hybrid technique may be used in which the clinician begins obturating the RCS by a lateral compaction technique. When the master cone and several accessory cones have been placed so that the mass is firmly lodged in the apical portion of the canal, a hot plugger is introduced, searing the points off approximately



Fig. 9.39 Obtura III unit with silver tips, GP plugs, and cleaning solution (Obtura Spartan, Earth City, MO).



Fig. 9.40 The Calamus thermoplastic unit (DENTSPLY Tulsa Dental Specialties, Tulsa, OK) for heating and injecting GP.

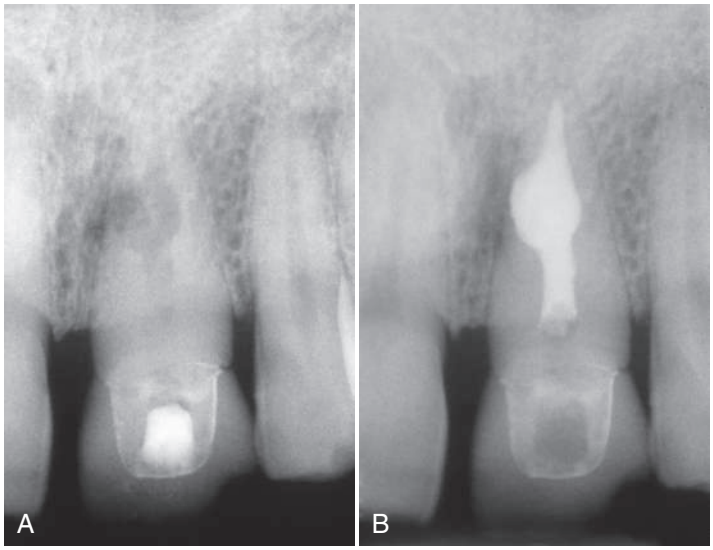


Fig. 9.38 Thermoplastic techniques are often used in cases with significant canal irregularities. **A,** A pretreatment radiograph of a maxillary central incisor exhibiting internal resorption. **B,** Posttreatment radiograph demonstrates a dense obturation of the resorptive defect with GP.



Fig. 9.41 The Elements obturation unit (Kerr Endo, Orange, CA) for injecting and compacting GP. Note the System B heat source.



Fig. 9.42 The battery-powered HotShot unit (Discus Dental, Culver City, CA) for heating and injecting GP.



Fig. 9.43 The Ultrafil 3D system consists of an injection syringe, GP cannulas, and heating unit (Coltène/Whaledent, Cuyahoga Falls, OH).

4 to 5 mm from the apex. Light vertical compaction is applied to restore the integrity of the apical plug of GP. The remainder of the canal is then filled with thermoplasticized GP injected as previously described, and the risk of overextension of GP is reduced.

Investigators studied, at 3, 6, and 12 months posttreatment, the healing rate of 236 teeth obturated with the Obtura system. Results indicated that 96% of the cases healed with the highest healing rate being in teeth filled flush with the radiographic apex (97%) when compared to overextensions (93%) and filling short (93%).[228]

Ultrafil 3D

Ultrafil 3D (Coltène/Whaledent) is a thermoplastic GP injection technique involving GP cannulas, a heating unit, and an injection syringe (see Fig. 9.43). The system employs three types of GP cannulas. Regular Set is a low-viscosity material that requires 30 minutes to set. Firm Set is also a low-viscosity material but differs in that it sets in 4 minutes. The manufacturer recommends compaction after the initial set with both materials. Endoset has a higher viscosity and does not flow as well. It is recommended for techniques employing compaction and sets in 2 minutes. The heater is preset at 90°C and does not require adjustment.

Each cannula has a 22-gauge stainless steel needle that measures 21 mm in length. The needles can be precurved. Cannulas can be disinfected but are not designed for heat sterilization procedures. Heating time varies but, for a cold unit, it takes 10 to 15 minutes. In a warm heater the recommended time is 3 minutes. After removing the cannula from the heater, the needle should be placed on the hot part of the heater for several seconds. The GP remains able to flow for 45 to 60 seconds depending on the viscosity.

Calamus

The Calamus flow obturation delivery system (DENTSPLY Sirona) is a thermoplastic device equipped with a cartridge system with 20- and 23-gauge needles (see Fig. 9.40). The unit permits control of temperature and the flow rate. Pluggers are also available for use with the system. The 360-degree activation switch allows great tactile sensation during use.

Elements

The Elements obturation unit (Kerr Endodontics) consists of a System B heat source and plugger as well as a hand piece extruder for delivering thermoplastic GP or RealSeal from a disposable cartridge (see Fig. 9.42). The cartridges come with 20-, 23-, and 25-gauge needles for GP and 20- and 23-gauges for RealSeal.

Hotshot

The HotShot delivery system (Discus Dental) is a cordless thermoplastic device that has a heating range from 150°C to 230°C (see Fig. 9.43). The unit is cordless and can be used with GP. Needles are available in 20, 23, and 25 gauges.

CARRIER-BASED GUTTA-PERCHA

Thermafil, Profile GT Obturators, GT Series X Obturators, and ProTaper Universal Obturators

Guttacore and Thermafil (DENTSPLY Sirona Dental Specialties) were introduced as GP obturation materials with a solid core (Fig. 9.44). Originally manufactured with a metal core and a coating of GP, the carrier was heated over an open flame. The technique was popular because the central core provided a rigid mechanism to facilitate the placement of the GP. Advantages included ease of placement and the



Fig. 9.44 Thermafil carrier and size verifier (DENTSPLY Sirona, Tulsa, OK).



Fig. 9.45 GT obturator and instrument (DENTSPLY Sirona, Tulsa, OK).

pliable properties of the GP. Disadvantages were that the metallic core made placement of a post challenging and that retreatment procedures were difficult. Additionally, the GP was often stripped from the carrier, leaving the carrier as the obturating material in the apical area of the canal. Reports have stated that carrier-based systems resulted in minimal gaps and voids in the resulting obturation mass.[325] Other reports have stated that there were no differences in healing rates between carrier-based systems and lateral compaction of GP.[117]

Changes in the carrier systems include the development of a plastic core coated with α-phase GP (Fig. 9.45) and a heating device that controls the temperature (Fig. 9.46). Obturators are designed to correspond to ISO standard file sizes, variable tapered nickel titanium rotary files, and ProFile GT and GT Series X nickel titanium rotary files (DENTSPLY Sirona). Size verifiers are available to aid in selection of the appropriate carrier and should fit passively at the working length (see Fig. 9.45).

As with all techniques, sealer is required. Grossman formulation sealers and resin sealers consistent with AH-26 and AH Plus are acceptable. However, Tubli-Seal and Wach's paste are not recommended.

The carrier is placed in the heating device. When the carrier is heated to the appropriate temperature, the clinician has approximately 10 seconds to retrieve it and insert it into the RCS (Fig. 9.47). This is accomplished without rotation or twisting. Evidence suggests that the insertion rate affects the obturation. The fill length and obturation of irregularities increase with increasing insertion rates, and a rapid insertion rate enhances obturation.[165]

Vertical compaction of the coronal GP can then be accomplished. When necessary, GP can be added, heat softened, and compacted. An advantage to this technique is the



Fig. 9.46 The Thermafil oven with carrier in place (DENTSPLY Tulsa Dental Specialties, Tulsa, OK).

potential for movement of GP into the lateral and accessory canals (Fig. 9.48);[308] however, extrusion of material beyond the apical extent of the root anatomical apex is a disadvantage.[154,197]

Pro-Post drills (DENTSPLY Sirona) are recommended if post space is required for restoration of the tooth. The unique eccentric cutting tip keeps the instrument centered in the canal while friction softens and removes the GP and plastic carrier.

When retreatment is necessary, the plastic carrier has a groove along its length to provide a potential access point for placement of a file to remove the carrier. Chloroform and hand files can be used to remove the GP surrounding the carrier. Rotary #.04 and #.06 nickel titanium files may

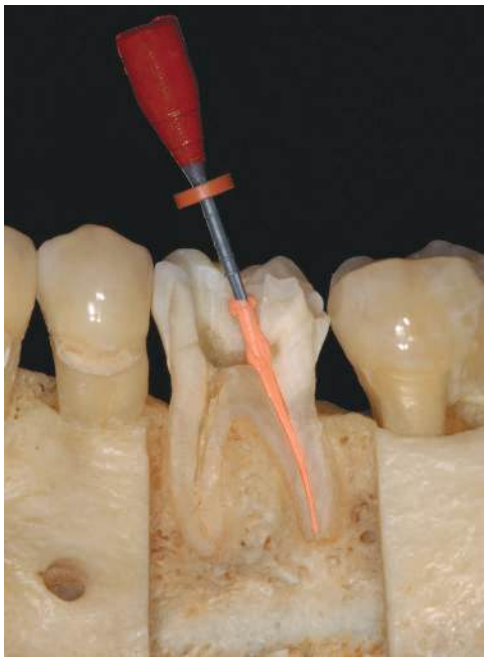


Fig. 9.47 Thermafil carrier placed in a distal canal.

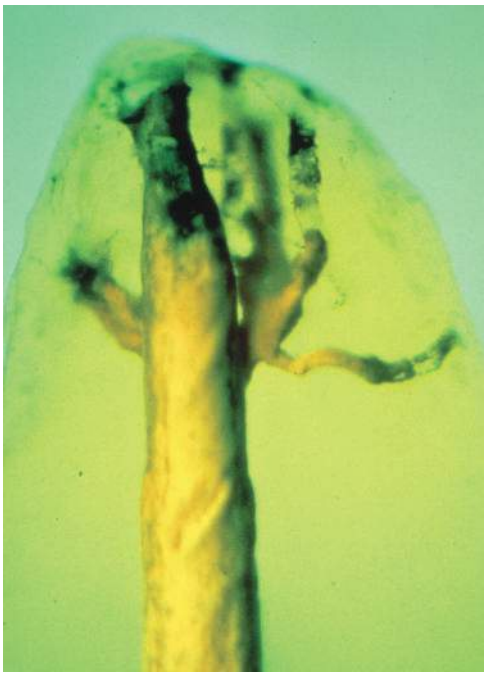


Fig. 9.48 Apical obturation of accessory canals by the Thermafil technique. (Courtesy DENTSPLY Sirona, Tulsa, OK. All rights owned by and used with permission from DENTSPLY International, Inc.)

also be used to remove the obturation materials. Retreatment rotary nickel titanium files are available in three different sizes to facilitate removal of GP and the carrier.

The plastic carriers are composed of two nontoxic materials. Sizes #20 to #40 are manufactured from a liquid crystal plastic. Sizes #40 to #90 are composed of polysulfone polymer. Both have similar physical characteristics, with the polysulfone carriers being susceptible to dissolution in chloroform.

GuttaCore (DENTSPLY Sirona), the latest generation of core carriers, uses cross-linked GP as the carrier of the outer thermoplasticized GP. This supposedly makes retreatment simpler as the clinician can simply drill through the carrier to gain access into the canal space. A recent in vitro study comparing the time required to remove GuttaCore, Thermafil Plus, and thermoplasticized GP from moderately curved RCS with ProTaper retreatment files reported that GuttaCore was removed in significantly less time when compared with the other two techniques.[25] The survival and healing rates after NS RCT with carrier-based obturation technique seem to be comparable to those previously reported for other obturation techniques.[223]

SuccessFil

SuccessFil (Coltène/Whaledent) is a carrier-based system associated with Ultrafil 3D (Fig. 9.49); however, the GP used in this technique comes in a syringe. Titanium or radiopaque plastic carriers are inserted into the syringe to the measured length of the canal. The GP is expressed on the carrier with the amount and shape determined by the rate of withdrawal from the syringe. Sealer is lightly coated on the canal walls, and the carrier with GP is placed into the RCS to the prepared length. The GP can be compacted around the carrier with various pluggers depending on the canal morphology. This is followed by severing of the carrier slightly above the orifice with a bur.

Simplifill

Simplifill (Kerr Endo) is GP manufactured for use after canal preparation with LightSpeed instruments (Fig. 9.50). The carrier has an apical plug with 5 mm of GP. The technique involves fitting a carrier that is consistent with the master apical rotary file to within 1 to 3 mm of the prepared length (Fig. 9.51). The apical GP plug can be modified by clipping the end in 1 mm increments to obtain an appropriate fit if the plug is too small. Once the cone is fitted, it is withdrawn and sealer applied to the canal walls. AH Plus is recommended. The Simplifill carrier is slowly advanced to the prepared length. This may require firm pressure. With the plug at the corrected working length, the handle is quickly



Fig. 9.49 SuccessFil is an additional carrier system (Coltène/Whaledent, Cuyahoga Falls, OH).

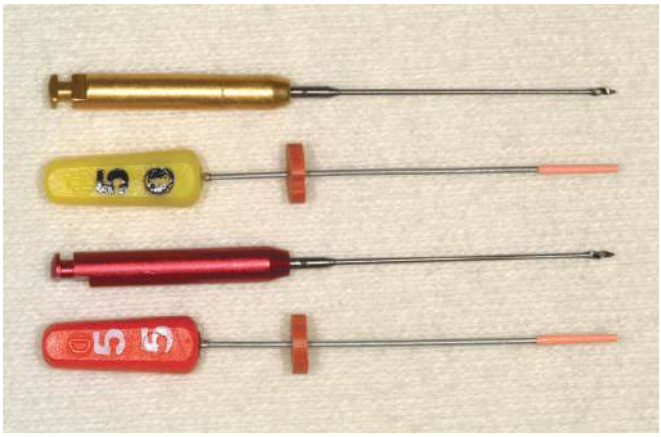


Fig. 9.50 Simplifill carrier and LightSpeed file (LightSpeed Technology, San Antonio, TX/Discus Dental, Culver City, CA).

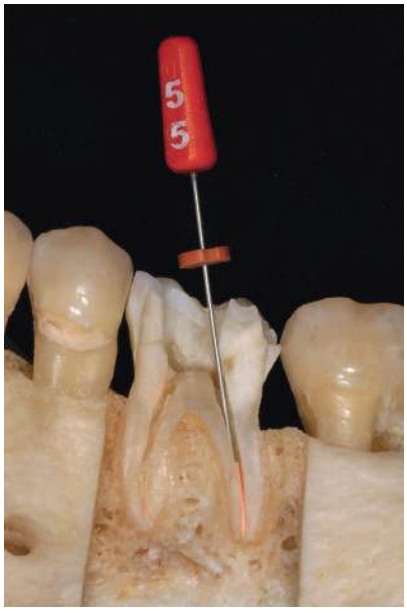


Fig. 9.51 Simplifill fitted to 1 to 3 mm from the prepared length.

rotated a minimum of four complete terms in a counterclockwise direction to separate the shaft from the apical GP. The coronal space can then be filled with GP using lateral compaction or a warm thermoplastic technique. When using lateral compaction, it is recommended that the first cone be the same size as the Simplifill carrier. This sectional technique is efficient and leakage potential is similar to that of other common techniques.[247]

THERMOMECHANICAL COMPACTION

McSpadden introduced the McSpadden Compactor, with flutes similar to a Hedström file, but in reverse. When activated in a slow-speed hand piece, the instrument would generate friction, soften the GP, and move it apically. Rotary compactors similar in design have been developed and advocated. To increase flexibility, the instrument is available in nickel titanium.

The technique requires fitting a GP master cone short of the prepared length and applying sealer. A compactor is selected on the basis of the size of the final RCS and inserted alongside the GP cone 3 to 4 mm from the prepared length. The hand piece is activated, and the friction of the rotating instrument heats the GP. The pliable mass is compacted apically and laterally as the device is withdrawn from the canal.

Advantages include simplicity of the armamentarium, the ability to fill canal irregularities,[251] and time. Disadvantages include possible extrusion of material, instrument fracture,[200] gouging canal walls, the inability to use the technique in curved canals, and possible excessive heat generation.[252] Microseal condensers (Kerr Dental) and the Gutta Condenser (DENTSPLY Sirona) are variations of this product.

SOLVENT TECHNIQUES

GP can be plasticized with solvents, for example, chloroform, eucalyptol, or xylol. Disadvantages include shrinkage caused by evaporation of the solvent, voids, the inability to achieve predictable length control of the obturating material, and irritation of periradicular tissues. Obturation techniques using solvents have been abandoned and replaced with materials and methods that exhibit minimum shrinkage.

PASTES

Pastes fulfill some of the criteria outlined by Grossman[110] and can adapt to the complex internal canal anatomy. However, the flow characteristic can result in extrusion or incomplete obturation. The inability to control the material is a distinct disadvantage, and, when extrusion occurs, it is possible that it can be corrected only by surgical intervention. In addition, pastes are sometimes used as a substitute for complete C&S procedures, and the addition of paraformaldehyde results in severe toxicity.

IMMEDIATE OBTURATION

Apical barriers may be necessary in cases with immature apical development, cases with external apical root resorption, and cases where instrumentation extends beyond the confines of the RCS. Dentin chips, $Ca(OH)_2$, demineralized dentin, lyophilized bone, tricalcium phosphate, hydroxyapatite, and collagen have been advocated for placement as a barrier in RCS exhibiting an open apex. The barriers are designed to permit obturation without extrusion of materials into the periradicular tissues but are often incomplete and do not seal the RCS.[242]

Dentin chips appear to confine materials to the RCS during instrumentation/obturation and may encourage development of a biologic seal.[242] A concern with this technique is contamination of the dentin with bacteria or other nonhost materials, because investigators found that infected dentin adversely affected healing.[208]

Calcium hydroxide has also been extensively used as a common apical barrier. $Ca(OH)_2$ has been shown to induce an apical barrier in apexification procedures. Calcifications similar to dentin plugs have been noted at the apical foramen.[208] Calcium hydroxide has the advantage of being free of bacterial contamination and may provide a better, although imperfect, apical seal.[303]

MTA (ProRoot MTA; DENTSPLY Sirona) has been successfully employed as an apical barrier material before obturation as an alternative to apexification, inducing cementum deposition and bone formation (Fig. 9.52).[264]

After C&S procedures, the RCS is dried and a small increment of MTA is placed to the radiographic apex and the location verified through a radiograph. If the material is overextended, it can be easily removed by irrigating with sterile saline. If it is short of the radiographic apex, it can be compacted with prefitted pluggers to move it to the radiographic apex. The material is compacted into the apical portion of the root to form a barrier. After the material sets, any thermoplasticized technique can be used to backfill with GP without concern of overextension (Fig. 9.53). Hand compaction of MTA is superior to direct placement of ultrasonic activation.[9] However, indirect ultrasonic placement of a 4 mm apical plug of MTA improved the seal.[317]

MTA is biocompatible and capable of inducing hard tissue formation.

Clinical evidence supports the use of bioactive endodontic cements like MTA.[292] In a study comparing $Ca(OH)_2$ with MTA in 15 children, each having paired immature teeth with necrotic pulps, two teeth treated with $Ca(OH)_2$ exhibited



Fig. 9.52 Mineral trioxide aggregate is available as ProRoot MTA. This material is advocated for use in apexification, repair of root perforations, repair of root resorption, root-end fillings, and pulp capping. (Courtesy DENTSPLY Tulsa Dental Specialties, Tulsa, OK, USA. All rights owned by and used with permission from DENTSPLY International Inc.)

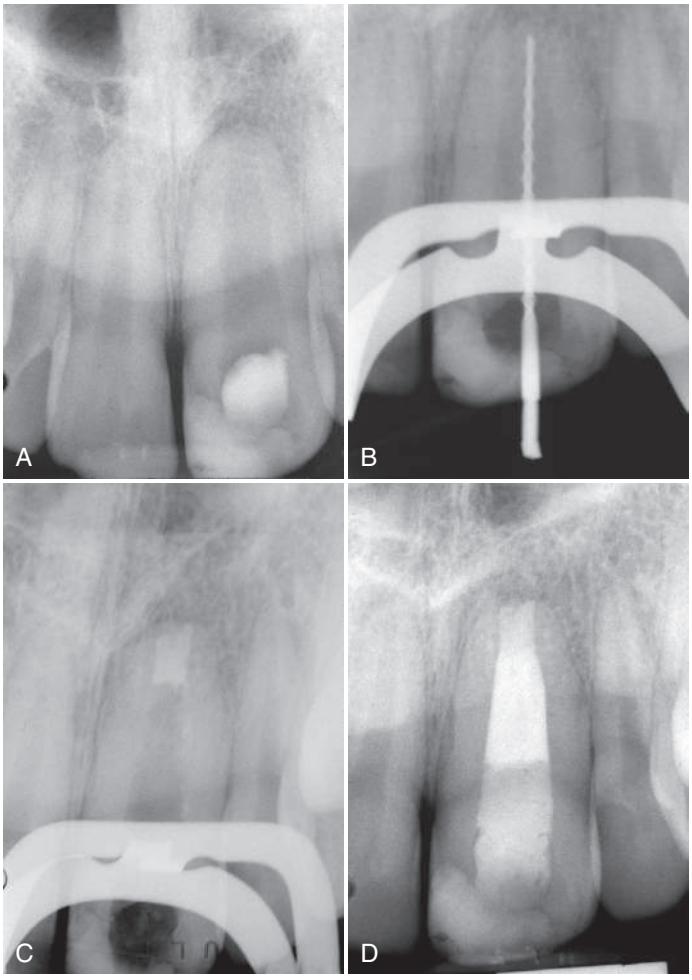


Fig. 9.53 Immediate obturation employs a barrier technique to prevent extrusion when the apex is open. This case involves a maxillary left central incisor with pulp necrosis caused by trauma. **A,** Pretreatment radiograph demonstrates a large canal with an open apex. **B,** Working length is established and the RCS is prepared. **C,** Mineral trioxide plug is placed. **D,** The canal is obturated with GP.

pathosis on recall examination whereas all the teeth treated with MTA were clinically and radiographically successful.[87] A prospective clinical study found the success rate of MTA barriers and immediate obturation in 43 cases to be 81%.[269] In another study, 85% of 20 teeth with immature apical development were considered healed and 5% were considered healing after immediate barrier placement.[127] The time required for healing is comparable to traditional apexification with $Ca(OH)_2$, and treatment time is reduced.[226]

Another option for treating immature teeth with pulp necrosis involves regenerative endodontic techniques (see Chapter 12). Advantages include continued increasing thickness of the canal walls, continued apical root development, and apical closure.

Coronal Orifice Seal

No matter which technique is used to obturate the RCS, coronal microleakage can occur within a short time through seemingly well-obturated canals,[293] potentially causing infection of the periapical area. Early research efforts focused on the quality of the seal in the apical part of the RCS to prevent percolation of apical fluids. However, contemporary research efforts have identified the greater importance of maintaining a coronal seal to prevent bacterial leakage.[137] Cavit (Premier Dental Products, 3M ESPE) has traditionally been advocated as an acceptable material. One study demonstrated that the placement of 3.5 mm of Cavit or Super EBA cement (Bosworth, Skokie, IL, USA) decreased bacterial leakage by 85% and 65%, respectively, when compared with unsealed controls, which all leaked at 45 days.[224] Another method to slow leakage through the obturated RCS, should failure of the coronal restoration occur, is to cover the floor of the pulp chamber with a lining of bonded material after the removal of excess GP and sealer to the canal oriface.[162] Investigators found that this procedure resulted in none of the experimental RCS demonstrating bacterial leakage at 60 days.[60] A material with a different color than dentin may be used for identification in case subsequent placement of a post is desired.

Future Technology and Nanodiamond-Embedded Gutta-Percha

Ongoing clinical trials are researching nanodiamond-embedded GP with some promising progress report and no adverse events.[159,160]

These studies might be the foundation for the continued clinical research of nanodiamond biomaterial and their application in endodontics.

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